BCHOICE - Simple Logit Model Example (Chocolate Candies) [Kuhfeld 2010] 10 subjects presented with 8 different chocolate candies choosing one Dark vs Milk Chocolate, Soft vs Chewy Center, Nuts vs No Nuts

In [5]:

```
title 'Conjoint Analysis of Chocolate Candies';

data Chocs;
   input Subj Choice Dark Soft Nuts;
   datalines;
1 0 0 0 0
1 0 0 0 1
1 0 0 1 0
1 0 0 1 1
1 1 1 0 0
1 0 1 0 1
1 0 1 1 0
1 0 1 1 1
2 0 0 0 0
2 0 0 0 1
2 0 0 1 0
2 0 0 1 1
2 0 1 0 0
2 1 1 0 1
2 0 1 1 0
2 0 1 1 1
3 0 0 0 0
3 0 0 0 1
3 0 0 1 0
3 0 0 1 1
3 0 1 0 0
3 0 1 0 1
3 1 1 1 0
3 0 1 1 1
4 0 0 0 0
4 0 0 0 1
4 0 0 1 0
4 0 0 1 1
4 1 1 0 0
4 0 1 0 1
4 0 1 1 0
4 0 1 1 1
5 0 0 0 0
5 1 0 0 1
5 0 0 1 0
5 0 0 1 1
5 0 1 0 0
5 0 1 0 1
5 0 1 1 0
5 0 1 1 1
6 0 0 0 0
6 0 0 0 1
6 0 0 1 0
6 0 0 1 1
6 0 1 0 0
6 1 1 0 1
6 0 1 1 0
6 0 1 1 1
```

```
7 0 0 0 0
7 1 0 0 1
7 0 0 1 0
7 0 0 1 1
7 0 1 0 0
7 0 1 0 1
7 0 1 1 0
7 0 1 1 1
8 0 0 0 0
8 0 0 0 1
8 0 0 1 0
8 0 0 1 1
8 0 1 0 0
8 1 1 0 1
8 0 1 1 0
8 0 1 1 1
9 0 0 0 0
9 0 0 0 1
9 0 0 1 0
9 0 0 1 1
9 0 1 0 0
9 1 1 0 1
9 0 1 1 0
9 0 1 1 1
10 0 0 0 0
10 0 0 0 1
10 0 0 1 0
10 0 0 1 1
10 0 1 0 0
10 1 1 0 1
10 0 1 1 0
10 0 1 1 1
;

proc print data=Chocs (obs=16);
run;
```

Conjoint Analysis of Chocolate Candies

Obs	Subj	Choice	Dark	Soft	Nuts
1	1	0	0	0	0
2	1	0	0	0	1
3	1	0	0	1	0
4	1	0	0	1	1
5	1	1	1	0	0
6	1	0	1	0	1
7	1	0	1	1	0
8	1	0	1	1	1
9	2	0	0	0	0
10	2	0	0	0	1
11	2	0	0	1	0

Obs	Subj	Choice	Dark	Soft	Nuts
12	2	0	0	1	1
13	2	0	1	0	0
14	2	1	1	0	1
15	2	0	1	1	0
16	2	0	1	1	1

In [6]:

```
ods graphics on;
proc bchoice data=Chocs outpost=Bsamp nmc=10000 thin=2 diag=(AutoCorr
     ESS MCSE) seed=124;
   class Dark(ref='0') Soft(ref='0') Nuts(ref='0') Subj;
   model Choice = Dark Soft Nuts / choiceset=(Subj) cprior=normal(var=1000);
run;
```

Conjoint Analysis of Chocolate Candies

The BCHOICE Procedure

Model Information	
Data Set	WORK.CHOCS
Response Variable	Choice
Type of Model	Logit
Type of Choice Response	Binary
Fixed Effects Included	Yes
Random Effects Included	No
Sampling Algorithm	Gamerman Metropolis
Burn-In Size	500
Simulation Size	10000
Thinning	2
Random Number Seed	124
Number of Threads	1

Class Level Information		
Class	Levels	Values
Dark	2	0 1
Soft	2	0 1
Nuts	2	0 1

Choice Sets Summary

Pattern	Choice Sets	Total Alternatives	Chosen Alternatives	Not Chosen

Choice Sets Summary

Pattern	Choice Sets	Total Alternatives	Chosen Alternatives	Not Chosen
1	10	8	1	7

Number of Observations	
Number of Observations Read	80
Number of Observations Used	80

Independent Normal Priors for Fixed Effects		
Parameter	Mean	Variance
Dark 1	0	1000
Soft 1	0	1000
Nuts 1	0	1000

Posterior Summaries and Intervals					
Parameter	N	Mean	Standard Deviation	95% HPD Interval	
Dark 1	5000	1.5308	0.7943	0.1848	3.2412
Soft 1	5000	-2.4312	0.9792	-4.4882	-0.8125
Nuts 1	5000	0.9671	0.7454	-0.3255	2.6675

Posterior Autocorrelations				
Parameter	Lag 1	Lag 5	Lag 10	Lag 50
Dark 1	0.4495	0.1427	0.0603	-0.0095
Soft 1	0.5389	0.1518	0.0730	0.0449
Nuts 1	0.4136	0.1473	0.0259	-0.0264

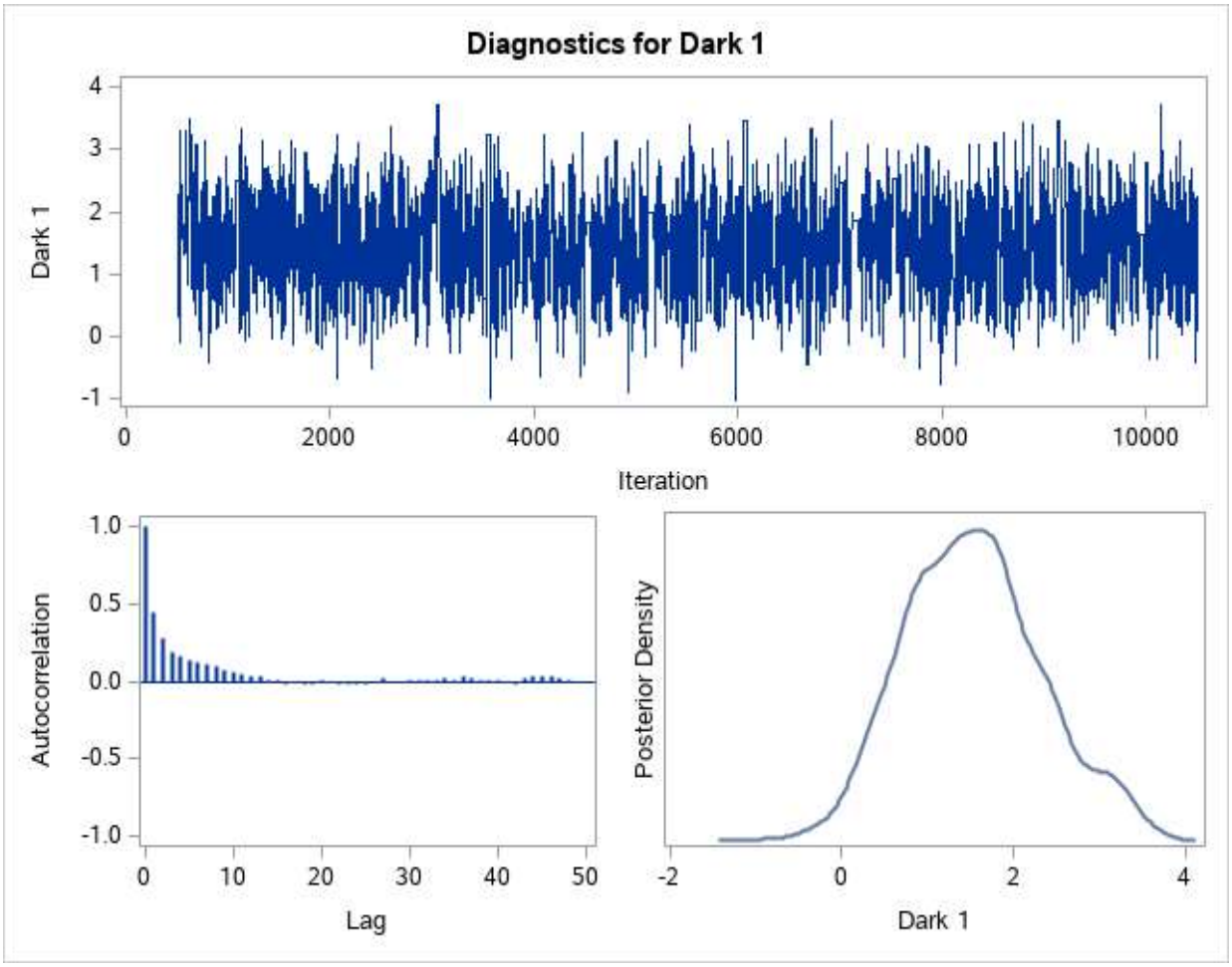
Effective Sample Sizes			
Parameter	ESS	Autocorrelation Time	Efficiency
Dark 1	1078.4	4.6365	0.2157
Soft 1	904.9	5.5253	0.1810
Nuts 1	1161.6	4.3043	0.2323

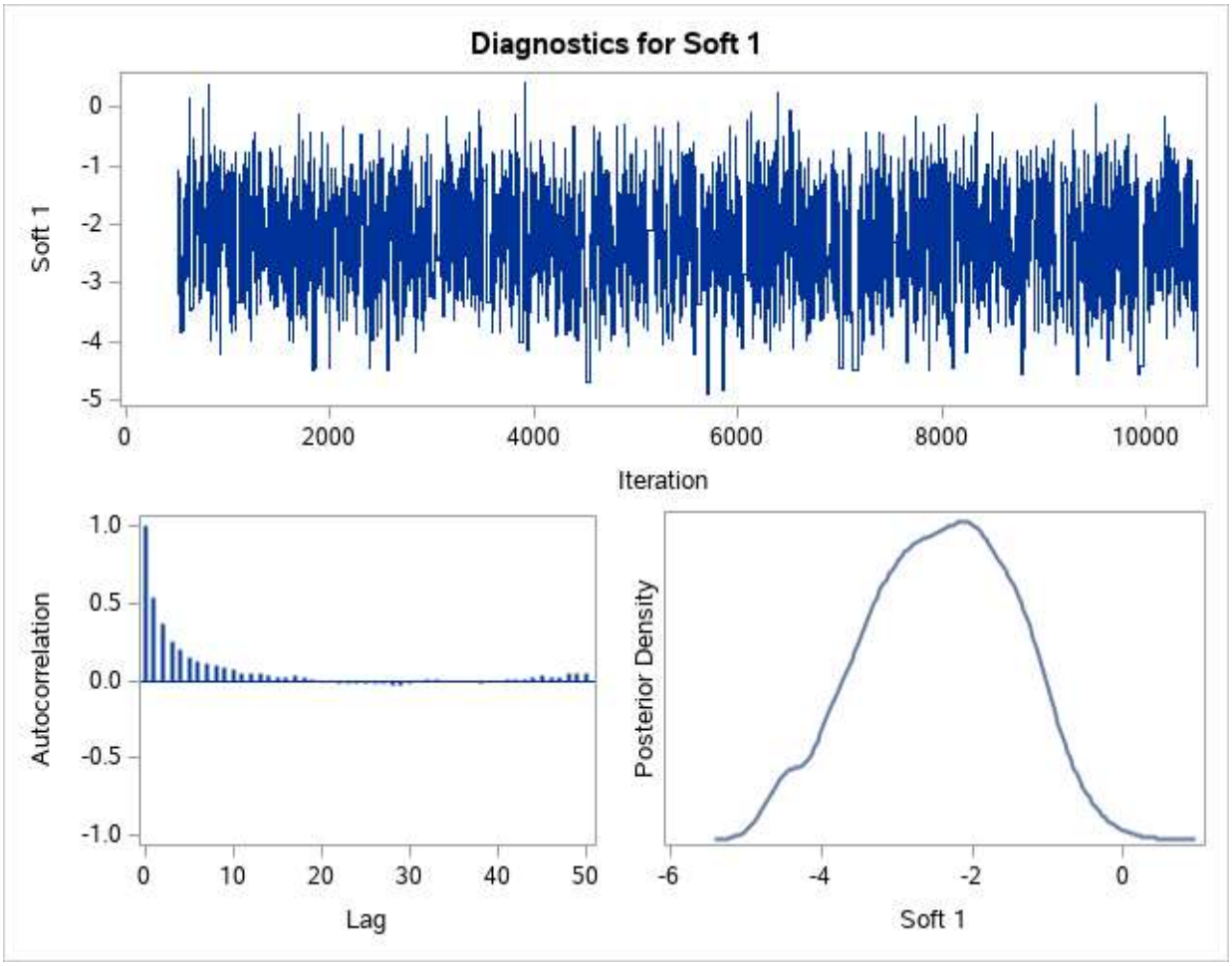
Monte Carlo Standard Errors			
Parameter	MCSE	Standard Deviation	MCSE/SD
Dark 1	0.0242	0.7943	0.0305

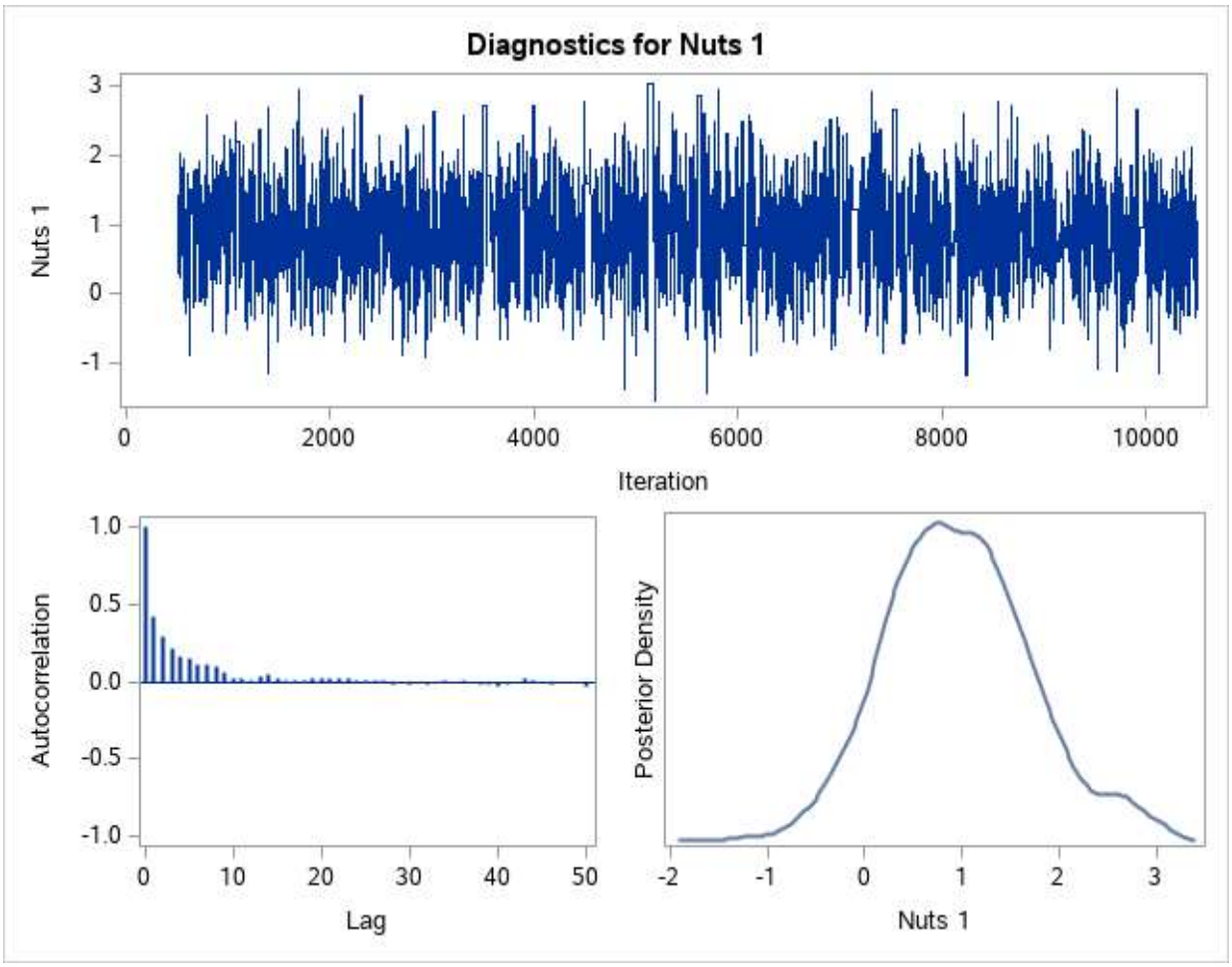
Monte Carlo Standard Errors			
Parameter	MCSE	Standard Deviation	MCSE/SD
Soft 1	0.0326	0.9792	0.0332
Nuts 1	0.0219	0.7454	0.0293

Conjoint Analysis of Chocolate Candies

The BCHOICE Procedure




Diagnostics for Soft 1
Soft 1
0
-1
-2
-3
-4
-5
0
2000
4000
6000
8000
10000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-6
-4
-2
0
Soft 1


Diagnostics for Nuts 1
Nuts 1
3
2
1
0
-1
0
2000
4000
6000
8000
10000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-2
-1
0
1
2
3
Nuts 1

BCHOICE - Logit Model with Random Effects Example (Trash Cans) [Kuhfeld 2010] 104 subjects presented with 10 choice tasks choosing one of each pairing Touchless opening, Material (steel/plastic), Automatic bag replace, Price ($40/80$)

In [7]:

```
title 'Random Coefficient Model TrashCans';

data Trashcan;
   input ID Task Choice Index Touchless Steel AutoBag Price80 @@;
   datalines;
1 1 1 1 0 1 1 0 1 1 0 2 1 1 0 0 1 2 0 1 0 0 0 0 1 2 1 2 1 1 1 1 1 3 0 1
0 0 0 0 1 3 1 2 1 1 0 0 1 4 0 1 0 1 0 1 1 4 1 2 1 0 0 1 1 5 1 1 0 1 1 0
1 5 0 2 1 0 0 1 1 6 0 1 0 0 1 1 1 6 1 2 1 1 1 1 1 7 0 1 1 0 0 1 1 7 1 2
1 1 0 0 1 8 0 1 0 0 1 1 1 8 1 2 0 1 1 0 1 9 1 1 0 0 1 1 1 9 0 2 1 0 0 1
1 10 0 1 0 1 1 0 1 10 1 2 1 0 1 0 2 1 1 1 0 1 1 0 2 1 0 2 1 1 0 0 2 2 0
1 0 0 0 0 2 2 1 2 1 1 1 1 2 3 0 1 0 0 0 0 2 3 1 2 1 1 0 0 2 4 0 1 0 1 0
1 2 4 1 2 1 0 0 1 2 5 1 1 0 1 1 0 2 5 0 2 1 0 0 1 2 6 1 1 0 0 1 1 2 6 0
2 1 1 1 1 2 7 1 1 1 0 0 1 2 7 0 2 1 1 0 0 2 8 1 1 0 0 1 1 2 8 0 2 0 1 1
0 2 9 1 1 0 0 1 1 2 9 0 2 1 0 0 1 2 10 1 1 0 1 1 0 2 10 0 2 1 0 1 0 3 1
1 1 0 1 1 0 3 1 0 2 1 1 0 0 3 2 0 1 0 0 0 0 3 2 1 2 1 1 1 1 3 3 0 1 0 0
0 0 3 3 1 2 1 1 0 0 3 4 1 1 0 1 0 1 3 4 0 2 1 0 0 1 3 5 0 1 0 1 1 0 3 5
1 2 1 0 0 1 3 6 0 1 0 0 1 1 3 6 1 2 1 1 1 1 3 7 0 1 1 0 0 1 3 7 1 2 1 1
0 0 3 8 0 1 0 0 1 1 3 8 1 2 0 1 1 0 3 9 0 1 0 0 1 1 3 9 1 2 1 0 0 1 3 10
1 1 0 1 1 0 3 10 0 2 1 0 1 0 4 1 1 1 0 1 1 0 4 1 0 2 1 1 0 0 4 2 1 1 0 0
0 0 4 2 0 2 1 1 1 1 4 3 0 1 0 0 0 0 4 3 1 2 1 1 0 0 4 4 1 1 0 1 0 1 4 4
0 2 1 0 0 1 4 5 1 1 0 1 1 0 4 5 0 2 1 0 0 1 4 6 0 1 0 0 1 1 4 6 1 2 1 1
1 1 4 7 0 1 1 0 0 1 4 7 1 2 1 1 0 0 4 8 0 1 0 0 1 1 4 8 1 2 0 1 1 0 4 9
1 1 0 0 1 1 4 9 0 2 1 0 0 1 4 10 0 1 0 1 1 0 4 10 1 2 1 0 1 0 5 1 1 1 0
1 1 0 5 1 0 2 1 1 0 0 5 2 0 1 0 0 0 0 5 2 1 2 1 1 1 1 5 3 0 1 0 0 0 0 5
3 1 2 1 1 0 0 5 4 1 1 0 1 0 1 5 4 0 2 1 0 0 1 5 5 1 1 0 1 1 0 5 5 0 2 1
0 0 1 5 6 0 1 0 0 1 1 5 6 1 2 1 1 1 1 5 7 0 1 1 0 0 1 5 7 1 2 1 1 0 0 5
8 0 1 0 0 1 1 5 8 1 2 0 1 1 0 5 9 1 1 0 0 1 1 5 9 0 2 1 0 0 1 5 10 1 1 0
1 1 0 5 10 0 2 1 0 1 0 6 1 0 1 0 1 1 0 6 1 1 2 1 1 0 0 6 2 1 1 0 0 0 0 6
2 0 2 1 1 1 1 6 3 0 1 0 0 0 0 6 3 1 2 1 1 0 0 6 4 0 1 0 1 0 1 6 4 1 2 1
0 0 1 6 5 1 1 0 1 1 0 6 5 0 2 1 0 0 1 6 6 0 1 0 0 1 1 6 6 1 2 1 1 1 1 6
7 0 1 1 0 0 1 6 7 1 2 1 1 0 0 6 8 0 1 0 0 1 1 6 8 1 2 0 1 1 0 6 9 0 1 0
0 1 1 6 9 1 2 1 0 0 1 6 10 0 1 0 1 1 0 6 10 1 2 1 0 1 0 7 1 0 1 0 1 1 0
7 1 1 2 1 1 0 0 7 2 0 1 0 0 0 0 7 2 1 2 1 1 1 1 7 3 0 1 0 0 0 0 7 3 1 2
1 1 0 0 7 4 0 1 0 1 0 1 7 4 1 2 1 0 0 1 7 5 1 1 0 1 1 0 7 5 0 2 1 0 0 1
7 6 0 1 0 0 1 1 7 6 1 2 1 1 1 1 7 7 0 1 1 0 0 1 7 7 1 2 1 1 0 0 7 8 0 1
0 0 1 1 7 8 1 2 0 1 1 0 7 9 0 1 0 0 1 1 7 9 1 2 1 0 0 1 7 10 0 1 0 1 1 0
7 10 1 2 1 0 1 0 8 1 1 1 0 1 1 0 8 1 0 2 1 1 0 0 8 2 0 1 0 0 0 0 8 2 1 2
1 1 1 1 8 3 1 1 0 0 0 0 8 3 0 2 1 1 0 0 8 4 1 1 0 1 0 1 8 4 0 2 1 0 0 1
8 5 1 1 0 1 1 0 8 5 0 2 1 0 0 1 8 6 1 1 0 0 1 1 8 6 0 2 1 1 1 1 8 7 1 1
1 0 0 1 8 7 0 2 1 1 0 0 8 8 1 1 0 0 1 1 8 8 0 2 0 1 1 0 8 9 1 1 0 0 1 1
8 9 0 2 1 0 0 1 8 10 1 1 0 1 1 0 8 10 0 2 1 0 1 0 9 1 1 1 0 1 1 0 9 1 0
2 1 1 0 0 9 2 0 1 0 0 0 0 9 2 1 2 1 1 1 1 9 3 0 1 0 0 0 0 9 3 1 2 1 1 0
0 9 4 0 1 0 1 0 1 9 4 1 2 1 0 0 1 9 5 1 1 0 1 1 0 9 5 0 2 1 0 0 1 9 6 0
1 0 0 1 1 9 6 1 2 1 1 1 1 9 7 0 1 1 0 0 1 9 7 1 2 1 1 0 0 9 8 0 1 0 0 1
1 9 8 1 2 0 1 1 0 9 9 1 1 0 0 1 1 9 9 0 2 1 0 0 1 9 10 1 1 0 1 1 0 9 10
0 2 1 0 1 0 10 1 1 1 0 1 1 0 10 1 0 2 1 1 0 0 10 2 0 1 0 0 0 0 10 2 1 2
1 1 1 1 10 3 0 1 0 0 0 0 10 3 1 2 1 1 0 0 10 4 0 1 0 1 0 1 10 4 1 2 1 0
0 1 10 5 1 1 0 1 1 0 10 5 0 2 1 0 0 1 10 6 1 1 0 0 1 1 10 6 0 2 1 1 1 1
10 7 0 1 1 0 0 1 10 7 1 2 1 1 0 0 10 8 0 1 0 0 1 1 10 8 1 2 0 1 1 0 10 9
1 1 0 0 1 1 10 9 0 2 1 0 0 1 10 10 1 1 0 1 1 0 10 10 0 2 1 0 1 0 11 1 1
1 0 1 1 0 11 1 0 2 1 1 0 0 11 2 0 1 0 0 0 0 11 2 1 2 1 1 1 1 11 3 0 1 0
0 0 0 11 3 1 2 1 1 0 0 11 4 0 1 0 1 0 1 11 4 1 2 1 0 0 1 11 5 1 1 0 1 1
```

0 11 5 0 2 1 0 0 1 11 6 0 1 0 0 1 1 11 6 1 2 1 1 1 1 11 7 0 1 1 0 0 1 11
7 1 2 1 1 0 0 11 8 0 1 0 0 1 1 11 8 1 2 0 1 1 0 11 9 1 1 0 0 1 1 11 9 0
2 1 0 0 1 11 10 0 1 0 1 1 0 11 10 1 2 1 0 1 0 12 1 1 1 0 1 1 0 12 1 0 2
1 1 0 0 12 2 1 1 0 0 0 0 12 2 0 2 1 1 1 1 12 3 1 1 0 0 0 0 12 3 0 2 1 1
0 0 12 4 1 1 0 1 0 1 12 4 0 2 1 0 0 1 12 5 1 1 0 1 1 0 12 5 0 2 1 0 0 1
12 6 0 1 0 0 1 1 12 6 1 2 1 1 1 1 12 7 0 1 1 0 0 1 12 7 1 2 1 1 0 0 12 8
0 1 0 0 1 1 12 8 1 2 0 1 1 0 12 9 1 1 0 0 1 1 12 9 0 2 1 0 0 1 12 10 1 1
0 1 1 0 12 10 0 2 1 0 1 0 13 1 1 1 0 1 1 0 13 1 0 2 1 1 0 0 13 2 0 1 0 0
0 0 13 2 1 2 1 1 1 1 13 3 0 1 0 0 0 0 13 3 1 2 1 1 0 0 13 4 1 1 0 1 0 1
13 4 0 2 1 0 0 1 13 5 1 1 0 1 1 0 13 5 0 2 1 0 0 1 13 6 0 1 0 0 1 1 13 6
1 2 1 1 1 1 13 7 0 1 1 0 0 1 13 7 1 2 1 1 0 0 13 8 0 1 0 0 1 1 13 8 1 2
0 1 1 0 13 9 1 1 0 0 1 1 13 9 0 2 1 0 0 1 13 10 1 1 0 1 1 0 13 10 0 2 1
0 1 0 14 1 0 1 0 1 1 0 14 1 1 2 1 1 0 0 14 2 1 1 0 0 0 0 14 2 0 2 1 1 1
1 14 3 0 1 0 0 0 0 14 3 1 2 1 1 0 0 14 4 1 1 0 1 0 1 14 4 0 2 1 0 0 1 14
5 1 1 0 1 1 0 14 5 0 2 1 0 0 1 14 6 0 1 0 0 1 1 14 6 1 2 1 1 1 1 14 7 0
1 1 0 0 1 14 7 1 2 1 1 0 0 14 8 0 1 0 0 1 1 14 8 1 2 0 1 1 0 14 9 0 1 0
0 1 1 14 9 1 2 1 0 0 1 14 10 1 1 0 1 1 0 14 10 0 2 1 0 1 0 15 1 0 1 0 1
1 0 15 1 1 2 1 1 0 0 15 2 1 1 0 0 0 0 15 2 0 2 1 1 1 1 15 3 0 1 0 0 0 0
15 3 1 2 1 1 0 0 15 4 0 1 0 1 0 1 15 4 1 2 1 0 0 1 15 5 1 1 0 1 1 0 15 5
0 2 1 0 0 1 15 6 0 1 0 0 1 1 15 6 1 2 1 1 1 1 15 7 0 1 1 0 0 1 15 7 1 2
1 1 0 0 15 8 0 1 0 0 1 1 15 8 1 2 0 1 1 0 15 9 0 1 0 0 1 1 15 9 1 2 1 0
0 1 15 10 0 1 0 1 1 0 15 10 1 2 1 0 1 0 16 1 0 1 0 1 1 0 16 1 1 2 1 1 0
0 16 2 0 1 0 0 0 0 16 2 1 2 1 1 1 1 16 3 0 1 0 0 0 0 16 3 1 2 1 1 0 0 16
4 0 1 0 1 0 1 16 4 1 2 1 0 0 1 16 5 1 1 0 1 1 0 16 5 0 2 1 0 0 1 16 6 0
1 0 0 1 1 16 6 1 2 1 1 1 1 16 7 0 1 1 0 0 1 16 7 1 2 1 1 0 0 16 8 0 1 0
0 1 1 16 8 1 2 0 1 1 0 16 9 0 1 0 0 1 1 16 9 1 2 1 0 0 1 16 10 0 1 0 1 1
0 16 10 1 2 1 0 1 0 17 1 1 1 0 1 1 0 17 1 0 2 1 1 0 0 17 2 0 1 0 0 0 0
17 2 1 2 1 1 1 1 17 3 0 1 0 0 0 0 17 3 1 2 1 1 0 0 17 4 1 1 0 1 0 1 17 4
0 2 1 0 0 1 17 5 1 1 0 1 1 0 17 5 0 2 1 0 0 1 17 6 0 1 0 0 1 1 17 6 1 2
1 1 1 1 17 7 0 1 1 0 0 1 17 7 1 2 1 1 0 0 17 8 0 1 0 0 1 1 17 8 1 2 0 1
1 0 17 9 1 1 0 0 1 1 17 9 0 2 1 0 0 1 17 10 1 1 0 1 1 0 17 10 0 2 1 0 1
0 18 1 1 1 0 1 1 0 18 1 0 2 1 1 0 0 18 2 0 1 0 0 0 0 18 2 1 2 1 1 1 1 18
3 0 1 0 0 0 0 18 3 1 2 1 1 0 0 18 4 1 1 0 1 0 1 18 4 0 2 1 0 0 1 18 5 1
1 0 1 1 0 18 5 0 2 1 0 0 1 18 6 0 1 0 0 1 1 18 6 1 2 1 1 1 1 18 7 0 1 1
0 0 1 18 7 1 2 1 1 0 0 18 8 0 1 0 0 1 1 18 8 1 2 0 1 1 0 18 9 1 1 0 0 1
1 18 9 0 2 1 0 0 1 18 10 0 1 0 1 1 0 18 10 1 2 1 0 1 0 19 1 1 1 0 1 1 0
19 1 0 2 1 1 0 0 19 2 1 1 0 0 0 0 19 2 0 2 1 1 1 1 19 3 0 1 0 0 0 0 19 3
1 2 1 1 0 0 19 4 0 1 0 1 0 1 19 4 1 2 1 0 0 1 19 5 1 1 0 1 1 0 19 5 0 2
1 0 0 1 19 6 0 1 0 0 1 1 19 6 1 2 1 1 1 1 19 7 0 1 1 0 0 1 19 7 1 2 1 1
0 0 19 8 0 1 0 0 1 1 19 8 1 2 0 1 1 0 19 9 1 1 0 0 1 1 19 9 0 2 1 0 0 1
19 10 0 1 0 1 1 0 19 10 1 2 1 0 1 0 20 1 1 1 0 1 1 0 20 1 0 2 1 1 0 0 20
2 1 1 0 0 0 0 20 2 0 2 1 1 1 1 20 3 0 1 0 0 0 0 20 3 1 2 1 1 0 0 20 4 0
1 0 1 0 1 20 4 1 2 1 0 0 1 20 5 1 1 0 1 1 0 20 5 0 2 1 0 0 1 20 6 0 1 0
0 1 1 20 6 1 2 1 1 1 1 20 7 0 1 1 0 0 1 20 7 1 2 1 1 0 0 20 8 0 1 0 0 1
1 20 8 1 2 0 1 1 0 20 9 1 1 0 0 1 1 20 9 0 2 1 0 0 1 20 10 1 1 0 1 1 0
20 10 0 2 1 0 1 0 21 1 0 1 0 1 1 0 21 1 1 2 1 1 0 0 21 2 1 1 0 0 0 0 21
2 0 2 1 1 1 1 21 3 0 1 0 0 0 0 21 3 1 2 1 1 0 0 21 4 1 1 0 1 0 1 21 4 0
2 1 0 0 1 21 5 1 1 0 1 1 0 21 5 0 2 1 0 0 1 21 6 0 1 0 0 1 1 21 6 1 2 1
1 1 1 21 7 0 1 1 0 0 1 21 7 1 2 1 1 0 0 21 8 0 1 0 0 1 1 21 8 1 2 0 1 1
0 21 9 0 1 0 0 1 1 21 9 1 2 1 0 0 1 21 10 1 1 0 1 1 0 21 10 0 2 1 0 1 0
22 1 0 1 0 1 1 0 22 1 1 2 1 1 0 0 22 2 0 1 0 0 0 0 22 2 1 2 1 1 1 1 22 3
0 1 0 0 0 0 22 3 1 2 1 1 0 0 22 4 1 1 0 1 0 1 22 4 0 2 1 0 0 1 22 5 1 1
0 1 1 0 22 5 0 2 1 0 0 1 22 6 0 1 0 0 1 1 22 6 1 2 1 1 1 1 22 7 0 1 1 0
0 1 22 7 1 2 1 1 0 0 22 8 0 1 0 0 1 1 22 8 1 2 0 1 1 0 22 9 0 1 0 0 1 1
22 9 1 2 1 0 0 1 22 10 0 1 0 1 1 0 22 10 1 2 1 0 1 0 23 1 1 1 0 1 1 0 23
1 0 2 1 1 0 0 23 2 0 1 0 0 0 0 23 2 1 2 1 1 1 1 23 3 0 1 0 0 0 0 23 3 1
2 1 1 0 0 23 4 1 1 0 1 0 1 23 4 0 2 1 0 0 1 23 5 1 1 0 1 1 0 23 5 0 2 1
0 0 1 23 6 1 1 0 0 1 1 23 6 0 2 1 1 1 1 23 7 0 1 1 0 0 1 23 7 1 2 1 1 0
0 23 8 0 1 0 0 1 1 23 8 1 2 0 1 1 0 23 9 1 1 0 0 1 1 23 9 0 2 1 0 0 1 23
10 1 1 0 1 1 0 23 10 0 2 1 0 1 0 24 1 1 1 0 1 1 0 24 1 0 2 1 1 0 0 24 2

1 1 0 0 0 0 24 2 0 2 1 1 1 1 24 3 1 1 0 0 0 0 24 3 0 2 1 1 0 0 24 4 1 1
0 1 0 1 24 4 0 2 1 0 0 1 24 5 1 1 0 1 1 0 24 5 0 2 1 0 0 1 24 6 1 1 0 0
1 1 24 6 0 2 1 1 1 1 24 7 0 1 1 0 0 1 24 7 1 2 1 1 0 0 24 8 0 1 0 0 1 1
24 8 1 2 0 1 1 0 24 9 1 1 0 0 1 1 24 9 0 2 1 0 0 1 24 10 1 1 0 1 1 0 24
10 0 2 1 0 1 0 25 1 0 1 0 1 1 0 25 1 1 2 1 1 0 0 25 2 0 1 0 0 0 0 25 2 1
2 1 1 1 1 25 3 0 1 0 0 0 0 25 3 1 2 1 1 0 0 25 4 0 1 0 1 0 1 25 4 1 2 1
0 0 1 25 5 0 1 0 1 1 0 25 5 1 2 1 0 0 1 25 6 0 1 0 0 1 1 25 6 1 2 1 1 1
1 25 7 0 1 1 0 0 1 25 7 1 2 1 1 0 0 25 8 0 1 0 0 1 1 25 8 1 2 0 1 1 0 25
9 0 1 0 0 1 1 25 9 1 2 1 0 0 1 25 10 0 1 0 1 1 0 25 10 1 2 1 0 1 0 26 1
1 1 0 1 1 0 26 1 0 2 1 1 0 0 26 2 1 1 0 0 0 0 26 2 0 2 1 1 1 1 26 3 0 1
0 0 0 0 26 3 1 2 1 1 0 0 26 4 0 1 0 1 0 1 26 4 1 2 1 0 0 1 26 5 1 1 0 1
1 0 26 5 0 2 1 0 0 1 26 6 0 1 0 0 1 1 26 6 1 2 1 1 1 1 26 7 0 1 1 0 0 1
26 7 1 2 1 1 0 0 26 8 0 1 0 0 1 1 26 8 1 2 0 1 1 0 26 9 1 1 0 0 1 1 26 9
0 2 1 0 0 1 26 10 1 1 0 1 1 0 26 10 0 2 1 0 1 0 27 1 1 1 0 1 1 0 27 1 0
2 1 1 0 0 27 2 1 1 0 0 0 0 27 2 0 2 1 1 1 1 27 3 0 1 0 0 0 0 27 3 1 2 1
1 0 0 27 4 0 1 0 1 0 1 27 4 1 2 1 0 0 1 27 5 1 1 0 1 1 0 27 5 0 2 1 0 0
1 27 6 0 1 0 0 1 1 27 6 1 2 1 1 1 1 27 7 0 1 1 0 0 1 27 7 1 2 1 1 0 0 27
8 0 1 0 0 1 1 27 8 1 2 0 1 1 0 27 9 1 1 0 0 1 1 27 9 0 2 1 0 0 1 27 10 0
1 0 1 1 0 27 10 1 2 1 0 1 0 28 1 0 1 0 1 1 0 28 1 1 2 1 1 0 0 28 2 1 1 0
0 0 0 28 2 0 2 1 1 1 1 28 3 0 1 0 0 0 0 28 3 1 2 1 1 0 0 28 4 1 1 0 1 0
1 28 4 0 2 1 0 0 1 28 5 1 1 0 1 1 0 28 5 0 2 1 0 0 1 28 6 0 1 0 0 1 1 28
6 1 2 1 1 1 1 28 7 0 1 1 0 0 1 28 7 1 2 1 1 0 0 28 8 0 1 0 0 1 1 28 8 1
2 0 1 1 0 28 9 1 1 0 0 1 1 28 9 0 2 1 0 0 1 28 10 1 1 0 1 1 0 28 10 0 2
1 0 1 0 29 1 0 1 0 1 1 0 29 1 1 2 1 1 0 0 29 2 0 1 0 0 0 0 29 2 1 2 1 1
1 1 29 3 0 1 0 0 0 0 29 3 1 2 1 1 0 0 29 4 0 1 0 1 0 1 29 4 1 2 1 0 0 1
29 5 1 1 0 1 1 0 29 5 0 2 1 0 0 1 29 6 0 1 0 0 1 1 29 6 1 2 1 1 1 1 29 7
0 1 1 0 0 1 29 7 1 2 1 1 0 0 29 8 0 1 0 0 1 1 29 8 1 2 0 1 1 0 29 9 0 1
0 0 1 1 29 9 1 2 1 0 0 1 29 10 0 1 0 1 1 0 29 10 1 2 1 0 1 0 30 1 1 1 0
1 1 0 30 1 0 2 1 1 0 0 30 2 0 1 0 0 0 0 30 2 1 2 1 1 1 1 30 3 0 1 0 0 0
0 30 3 1 2 1 1 0 0 30 4 1 1 0 1 0 1 30 4 0 2 1 0 0 1 30 5 1 1 0 1 1 0 30
5 0 2 1 0 0 1 30 6 0 1 0 0 1 1 30 6 1 2 1 1 1 1 30 7 0 1 1 0 0 1 30 7 1
2 1 1 0 0 30 8 0 1 0 0 1 1 30 8 1 2 0 1 1 0 30 9 1 1 0 0 1 1 30 9 0 2 1
0 0 1 30 10 1 1 0 1 1 0 30 10 0 2 1 0 1 0 31 1 1 1 0 1 1 0 31 1 0 2 1 1
0 0 31 2 0 1 0 0 0 0 31 2 1 2 1 1 1 1 31 3 0 1 0 0 0 0 31 3 1 2 1 1 0 0
31 4 0 1 0 1 0 1 31 4 1 2 1 0 0 1 31 5 1 1 0 1 1 0 31 5 0 2 1 0 0 1 31 6
0 1 0 0 1 1 31 6 1 2 1 1 1 1 31 7 0 1 1 0 0 1 31 7 1 2 1 1 0 0 31 8 0 1
0 0 1 1 31 8 1 2 0 1 1 0 31 9 1 1 0 0 1 1 31 9 0 2 1 0 0 1 31 10 0 1 0 1
1 0 31 10 1 2 1 0 1 0 32 1 1 1 0 1 1 0 32 1 0 2 1 1 0 0 32 2 1 1 0 0 0 0
32 2 0 2 1 1 1 1 32 3 1 1 0 0 0 0 32 3 0 2 1 1 0 0 32 4 1 1 0 1 0 1 32 4
0 2 1 0 0 1 32 5 1 1 0 1 1 0 32 5 0 2 1 0 0 1 32 6 1 1 0 0 1 1 32 6 0 2
1 1 1 1 32 7 0 1 1 0 0 1 32 7 1 2 1 1 0 0 32 8 0 1 0 0 1 1 32 8 1 2 0 1
1 0 32 9 1 1 0 0 1 1 32 9 0 2 1 0 0 1 32 10 1 1 0 1 1 0 32 10 0 2 1 0 1
0 33 1 1 1 0 1 1 0 33 1 0 2 1 1 0 0 33 2 0 1 0 0 0 0 33 2 1 2 1 1 1 1 33
3 0 1 0 0 0 0 33 3 1 2 1 1 0 0 33 4 1 1 0 1 0 1 33 4 0 2 1 0 0 1 33 5 1
1 0 1 1 0 33 5 0 2 1 0 0 1 33 6 0 1 0 0 1 1 33 6 1 2 1 1 1 1 33 7 0 1 1
0 0 1 33 7 1 2 1 1 0 0 33 8 0 1 0 0 1 1 33 8 1 2 0 1 1 0 33 9 1 1 0 0 1
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0 83 8 0 1 0 0 1 1 83 8 1 2 0 1 1 0 83 9 1 1 0 0 1 1 83 9 0 2 1 0 0 1 83
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```
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8 1 2 0 1 1 0 100 9 0 1 0 0 1 1 100 9 1 2 1 0 0 1 100 10 1 1 0 1 1 0 100
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6 1 2 1 1 1 1 102 7 0 1 1 0 0 1 102 7 1 2 1 1 0 0 102 8 0 1 0 0 1 1 102
8 1 2 0 1 1 0 102 9 1 1 0 0 1 1 102 9 0 2 1 0 0 1 102 10 1 1 0 1 1 0 102
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6 1 2 1 1 1 1 103 7 0 1 1 0 0 1 103 7 1 2 1 1 0 0 103 8 0 1 0 0 1 1 103
8 1 2 0 1 1 0 103 9 1 1 0 0 1 1 103 9 0 2 1 0 0 1 103 10 0 1 0 1 1 0 103
10 1 2 1 0 1 0 104 1 1 1 0 1 1 0 104 1 0 2 1 1 0 0 104 2 0 1 0 0 0 0 104
2 1 2 1 1 1 1 104 3 0 1 0 0 0 0 104 3 1 2 1 1 0 0 104 4 0 1 0 1 0 1 104
4 1 2 1 0 0 1 104 5 1 1 0 1 1 0 104 5 0 2 1 0 0 1 104 6 0 1 0 0 1 1 104
6 1 2 1 1 1 1 104 7 0 1 1 0 0 1 104 7 1 2 1 1 0 0 104 8 0 1 0 0 1 1 104
8 1 2 0 1 1 0 104 9 0 1 0 0 1 1 104 9 1 2 1 0 0 1 104 10 0 1 0 1 1 0 104
10 1 2 1 0 1 0
;

proc print data=Trashcan (obs=8);
run;
```

Random Coefficient Model TrashCans

Obs	ID	Task	Choice	Index	Touchless	Steel	AutoBag	Price80
1	1	1	1	1	0	1	1	0
2	1	1	0	2	1	1	0	0
3	1	2	0	1	0	0	0	0
4	1	2	1	2	1	1	1	1
5	1	3	0	1	0	0	0	0
6	1	3	1	2	1	1	0	0
7	1	4	0	1	0	1	0	1
8	1	4	1	2	1	0	0	1

In [8]:

```
proc bchoice data=Trashcan seed=1 nmc=30000 thin=2 nthreads=4;
   class ID Task;
   model Choice = Touchless Steel AutoBag Price80 / choiceset=(ID Task);
   random Touchless Steel AutoBag Price80 / sub=ID monitor=(1 to 5) type=un;
run;
```

Random Coefficient Model TrashCans

The BCHOICE Procedure

Model Information	
Data Set	WORK.TRASHCAN
Response Variable	Choice
Type of Model	Logit
Type of Choice Response	Binary
Fixed Effects Included	Yes
Random Effects Included	Yes
Sampling Algorithm	Gamerman Metropolis
Burn-In Size	500
Simulation Size	30000
Thinning	2
Random Number Seed	1
Number of Threads	4

Choice Sets Summary				
Pattern	Choice Sets	Total Alternatives	Chosen Alternatives	Not Chosen
1	1040	2	1	1

Number of Observations	
Number of Observations Read	2080
Number of Observations Used	2080

Independent Normal Priors for Fixed Effects		
Parameter	Mean	Variance
Touchless	0	100
Steel	0	100
AutoBag	0	100
Price80	0	100

Priors for Covariance Parameters of Random Effects			
Parameter	Prior Distribution	DF	Scale
RECov	Inverse Wishart	7	7

Posterior Summaries and Intervals

Parameter	Subject	N	Mean	Standard Deviation	95% HPD Interval	
Touchless		15000	1.7043	0.2709	1.1645	2.2322
Steel		15000	1.0516	0.2608	0.5376	1.5691
AutoBag		15000	2.1722	0.3583	1.4886	2.8948
Price80		15000	-4.6321	0.6720	-5.9688	-3.4046
RECov Touchless, Touchless		15000	3.1145	1.0613	1.3441	5.2576
RECov Steel, Touchless		15000	-0.4727	0.8796	-2.1884	1.2334
RECov Steel, Steel		15000	2.6312	0.9565	1.1035	4.5817
RECov AutoBag, Touchless		15000	-0.6473	0.8811	-2.3803	1.1974
RECov AutoBag, Steel		15000	0.0595	0.7078	-1.4318	1.3748
RECov AutoBag, AutoBag		15000	3.5547	1.5837	0.9611	6.7248
RECov Price80, Touchless		15000	-1.3016	1.1857	-3.8046	0.7154
RECov Price80, Steel		15000	-1.5935	1.2052	-4.1428	0.6170
RECov Price80, AutoBag		15000	-2.2190	1.7156	-5.7073	0.7349
RECov Price80, Price80		15000	7.7452	3.4362	2.2287	14.3546
Touchless	ID 1	15000	0.5244	1.0525	-1.3366	2.7890
Steel	ID 1	15000	-0.3298	1.0799	-2.3057	1.9751
AutoBag	ID 1	15000	1.6216	1.3842	-0.9207	4.4860
Price80	ID 1	15000	-0.5394	2.2018	-5.2251	3.4324
Touchless	ID 2	15000	-1.7498	0.9341	-3.5965	0.0610
Steel	ID 2	15000	-1.4343	0.8572	-3.2176	0.2048
AutoBag	ID 2	15000	0.4767	1.3208	-2.0239	3.1743
Price80	ID 2	15000	4.6873	1.4955	1.9223	7.7535
Touchless	ID 3	15000	-0.6191	0.9695	-2.4024	1.4490
Steel	ID 3	15000	0.5779	0.9524	-1.2792	2.4187
AutoBag	ID 3	15000	-1.8737	1.2107	-4.2878	0.4407
Price80	ID 3	15000	3.7507	1.6747	0.4795	7.0672
Touchless	ID 4	15000	-0.5921	0.9679	-2.5222	1.2694
Steel	ID 4	15000	0.3152	1.0416	-1.7134	2.3286
AutoBag	ID 4	15000	0.9770	1.4250	-1.7103	3.9276
Price80	ID 4	15000	-1.9068	2.3717	-6.4256	2.8233
Touchless	ID 5	15000	-1.2689	1.1491	-3.4720	0.9828
Steel	ID 5	15000	1.6581	1.2220	-0.4731	4.2409

Posterior Summaries and Intervals

Parameter	Subject	N	Mean	Standard Deviation	95% HPD Interval	
AutoBag	ID 5	15000	1.2695	1.5708	-1.7506	4.4762
Price80	ID 5	15000	-0.3706	2.3810	-5.0593	4.2134

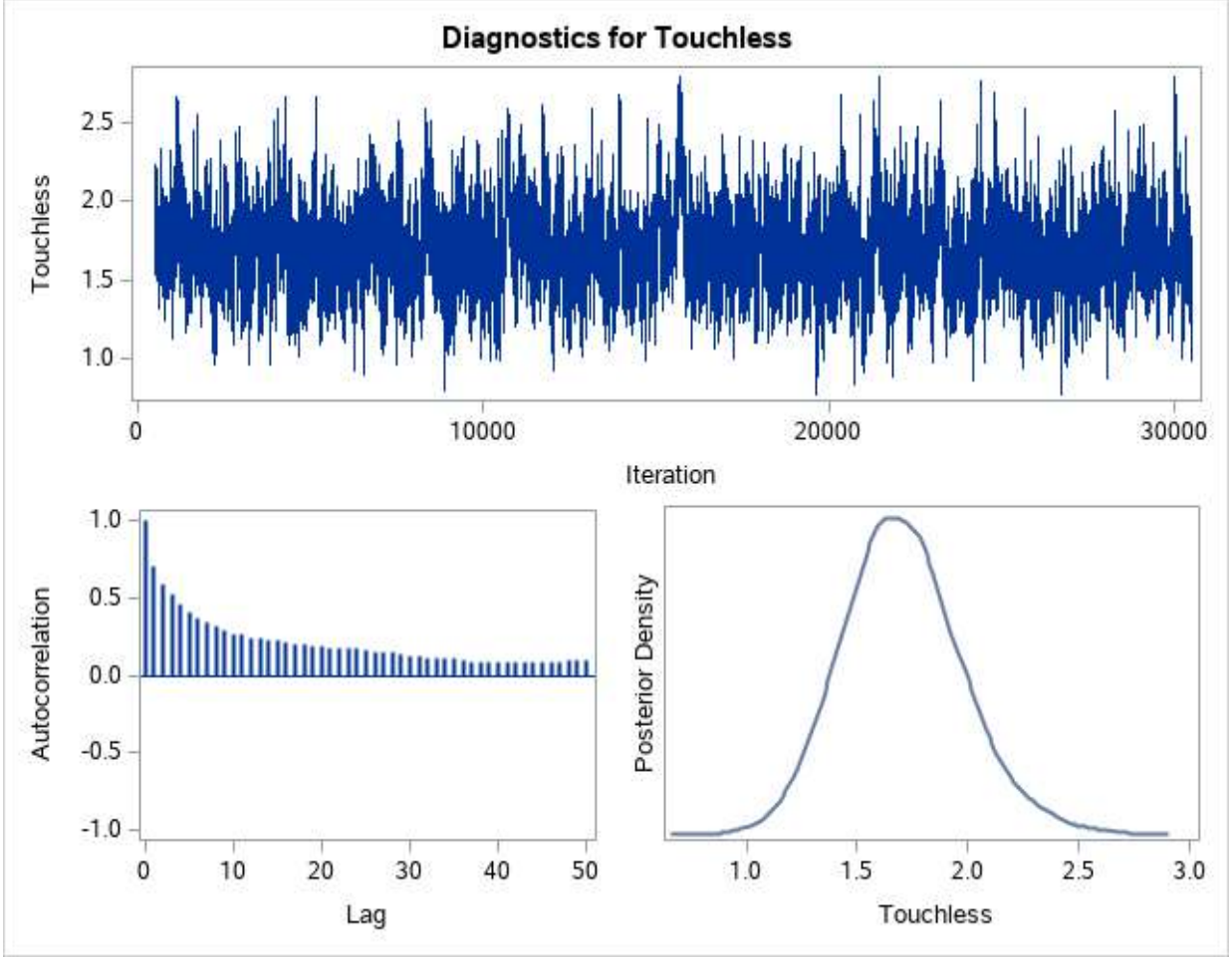
Effective Sample Sizes

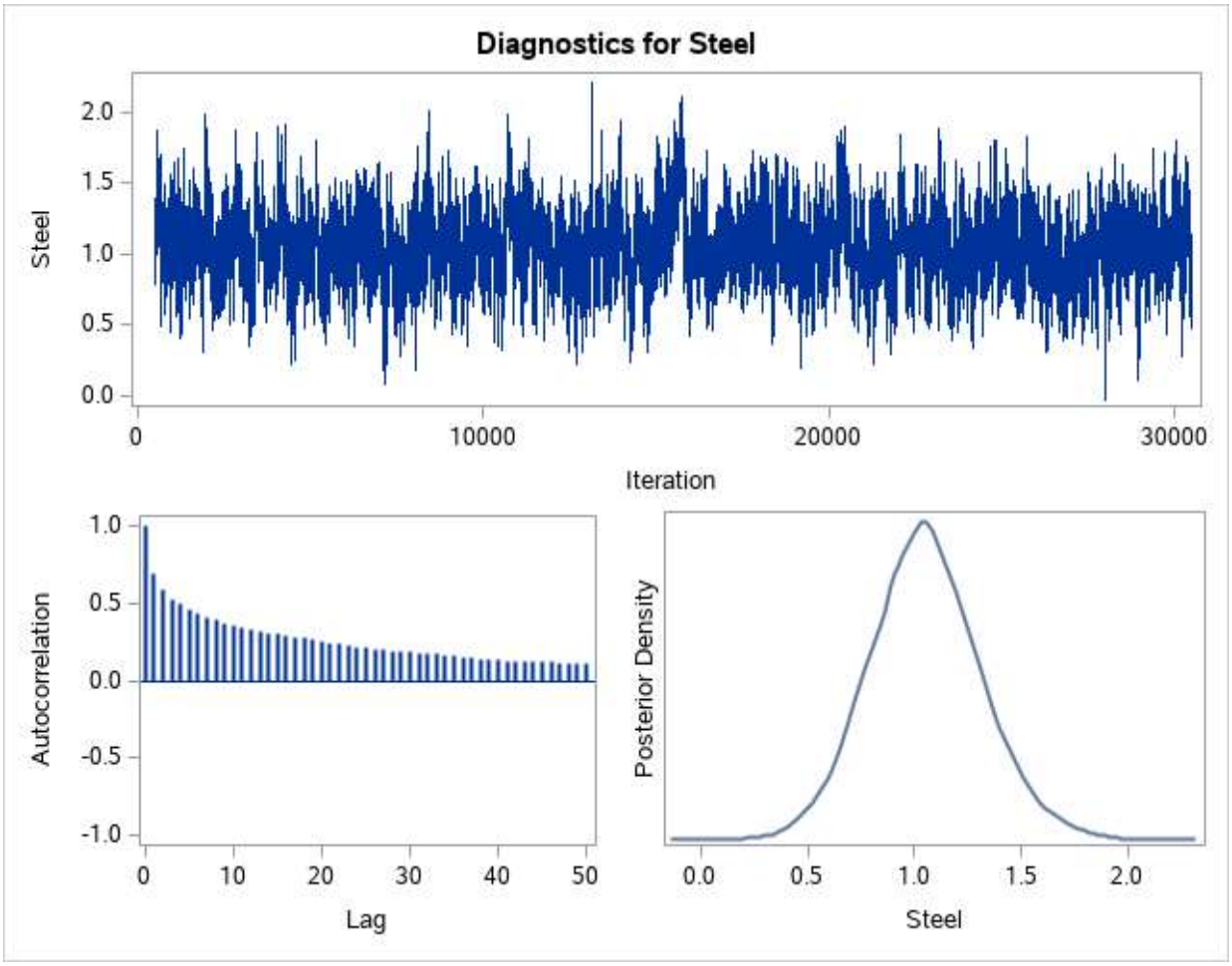
Parameter	Subject	ESS	Autocorrelation Time	Efficiency
Touchless		636.8	23.5556	0.0425
Steel		513.2	29.2262	0.0342
AutoBag		538.9	27.8356	0.0359
Price80		328.6	45.6444	0.0219
RECov Touchless, Touchless		411.2	36.4809	0.0274
RECov Steel, Touchless		311.0	48.2261	0.0207
RECov Steel, Steel		472.5	31.7427	0.0315
RECov AutoBag, Touchless		400.5	37.4510	0.0267
RECov AutoBag, Steel		518.0	28.9587	0.0345
RECov AutoBag, AutoBag		366.0	40.9884	0.0244
RECov Price80, Touchless		462.7	32.4166	0.0308
RECov Price80, Steel		386.5	38.8075	0.0258
RECov Price80, AutoBag		370.9	40.4438	0.0247
RECov Price80, Price80		309.8	48.4209	0.0207
Touchless	ID 1	3833.4	3.9130	0.2556
Steel	ID 1	4662.7	3.2171	0.3108
AutoBag	ID 1	2212.0	6.7813	0.1475
Price80	ID 1	2526.3	5.9376	0.1684
Touchless	ID 2	1844.5	8.1323	0.1230
Steel	ID 2	2346.6	6.3923	0.1564
AutoBag	ID 2	1616.5	9.2792	0.1078
Price80	ID 2	817.7	18.3435	0.0545
Touchless	ID 3	1191.3	12.5917	0.0794
Steel	ID 3	1854.9	8.0868	0.1237
AutoBag	ID 3	1147.1	13.0760	0.0765
Price80	ID 3	746.7	20.0873	0.0498

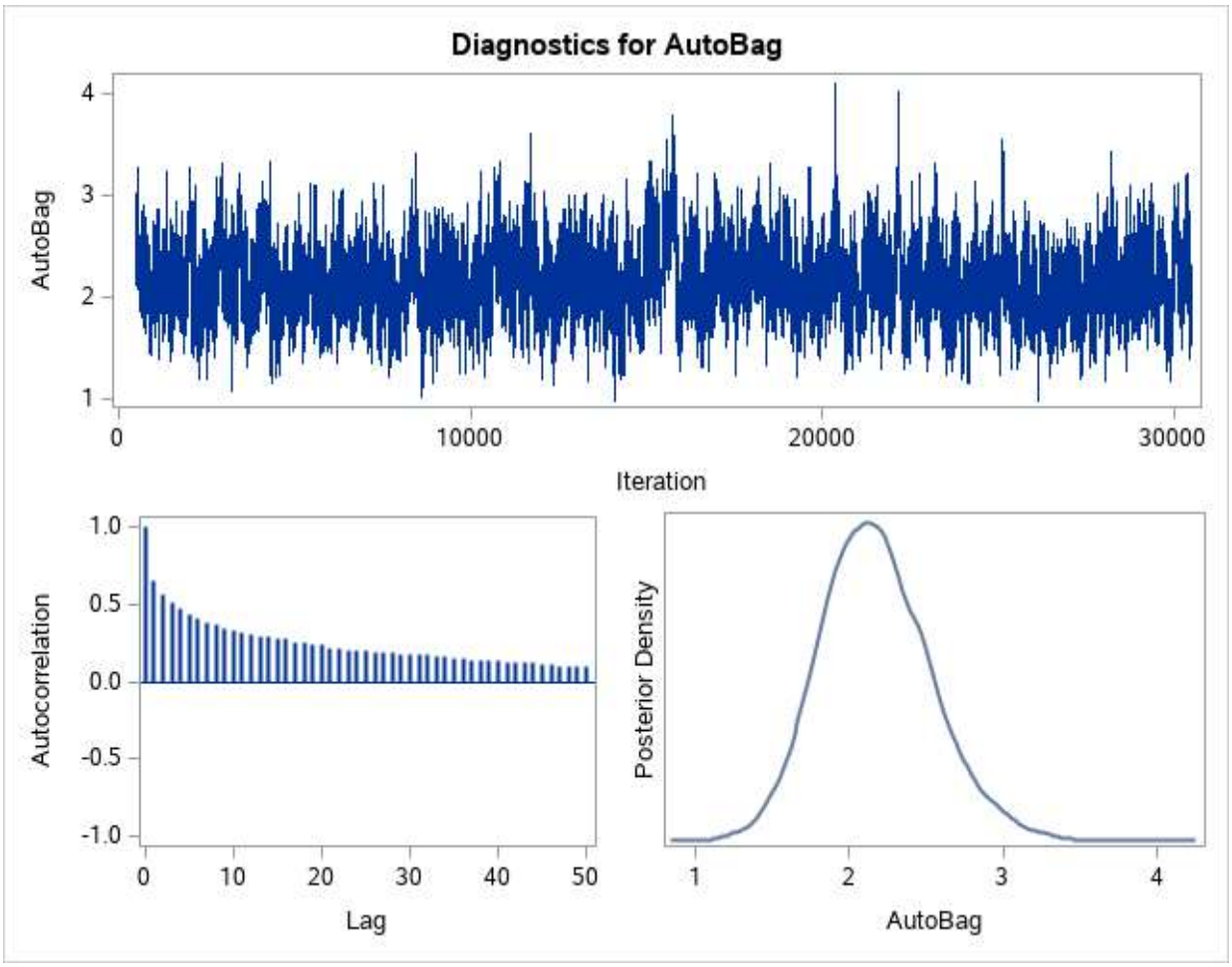
Effective Sample Sizes				
Parameter	Subject	ESS	Autocorrelation Time	Efficiency
Touchless	ID 4	6831.1	2.1958	0.4554
Steel	ID 4	6804.5	2.2044	0.4536
AutoBag	ID 4	3658.7	4.0998	0.2439
Price80	ID 4	4232.5	3.5440	0.2822
Touchless	ID 5	2329.9	6.4381	0.1553
Steel	ID 5	2180.3	6.8797	0.1454
AutoBag	ID 5	2136.4	7.0210	0.1424
Price80	ID 5	3241.3	4.6278	0.2161

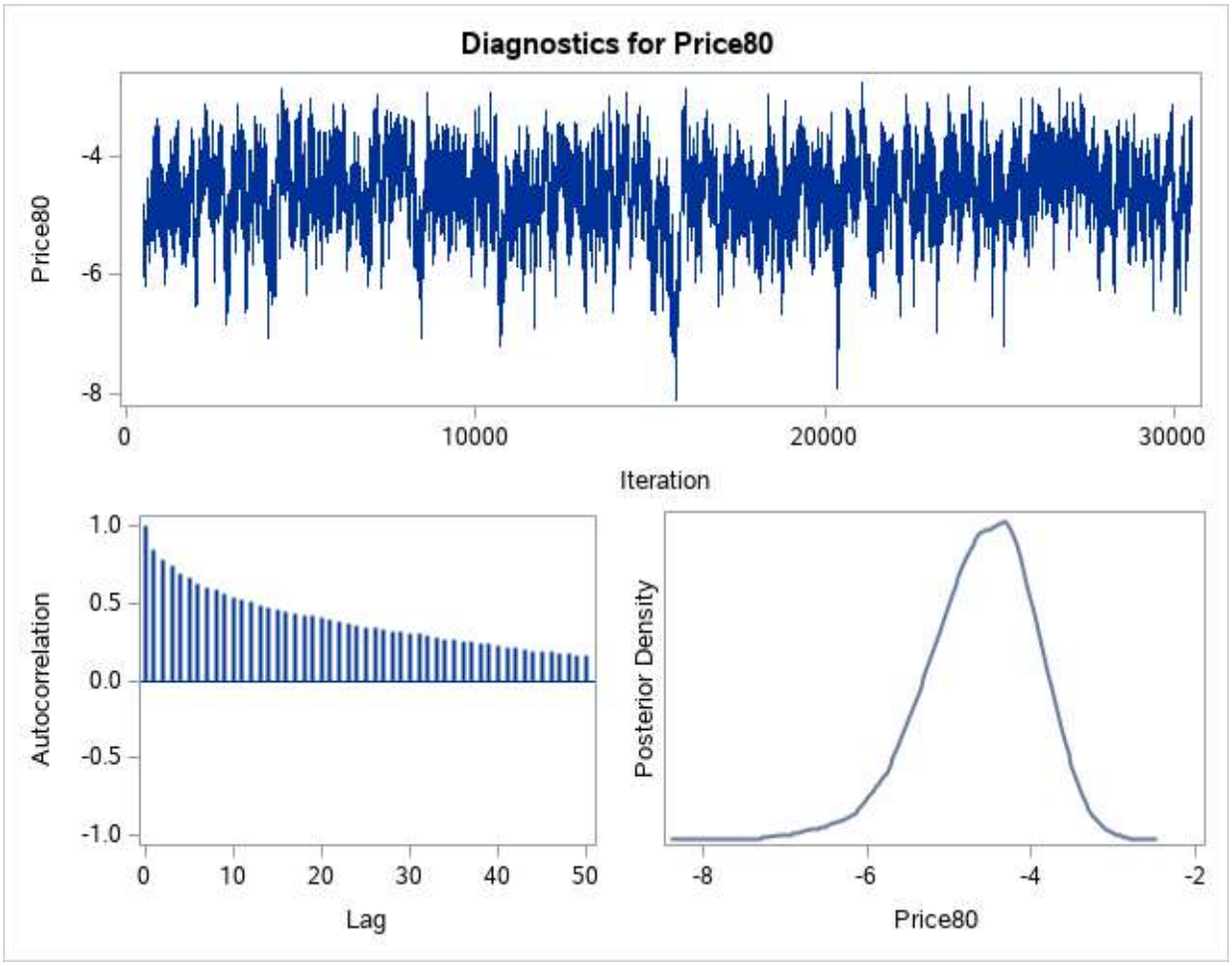
Random Coefficient Model TrashCans

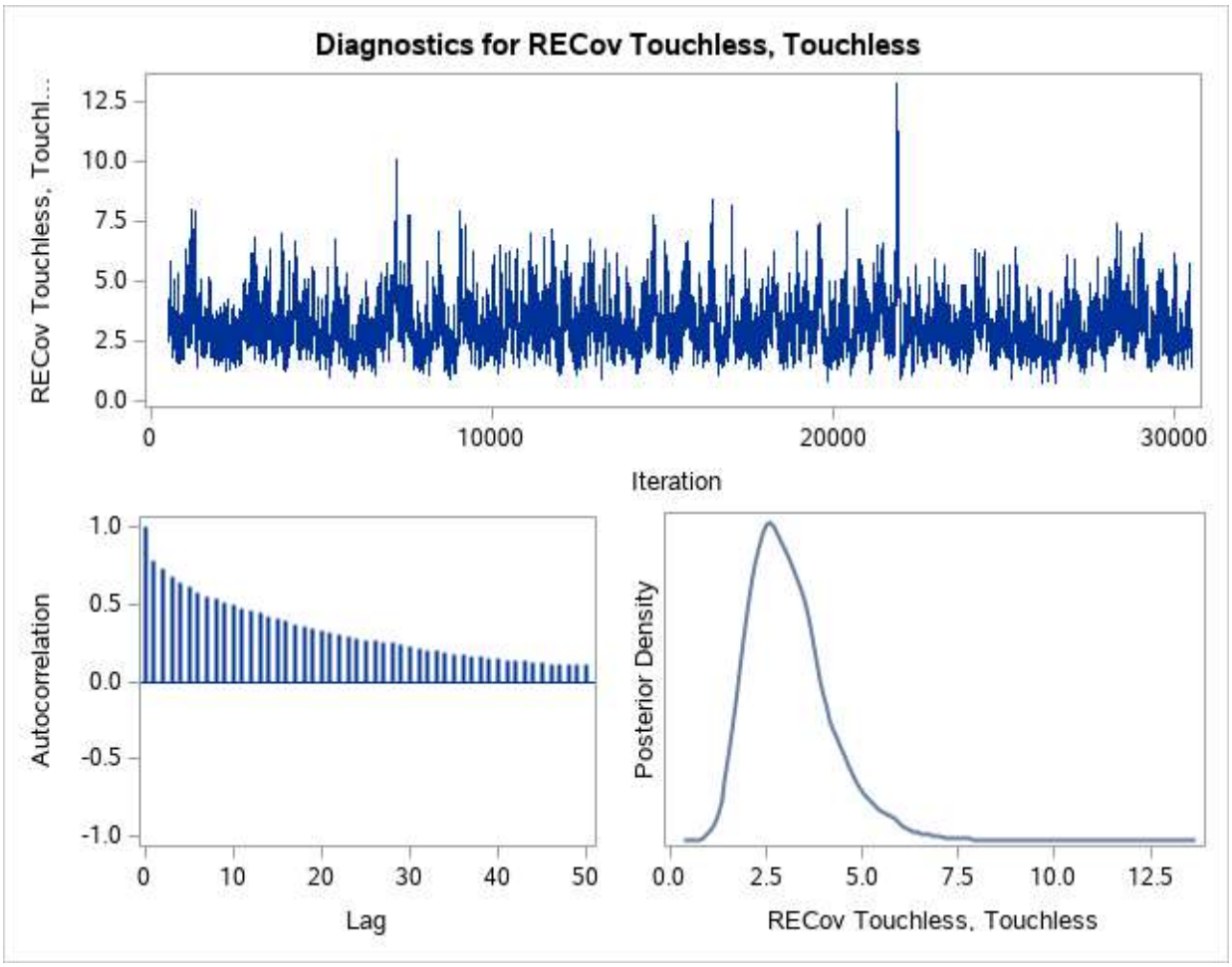
The BCHOICE Procedure

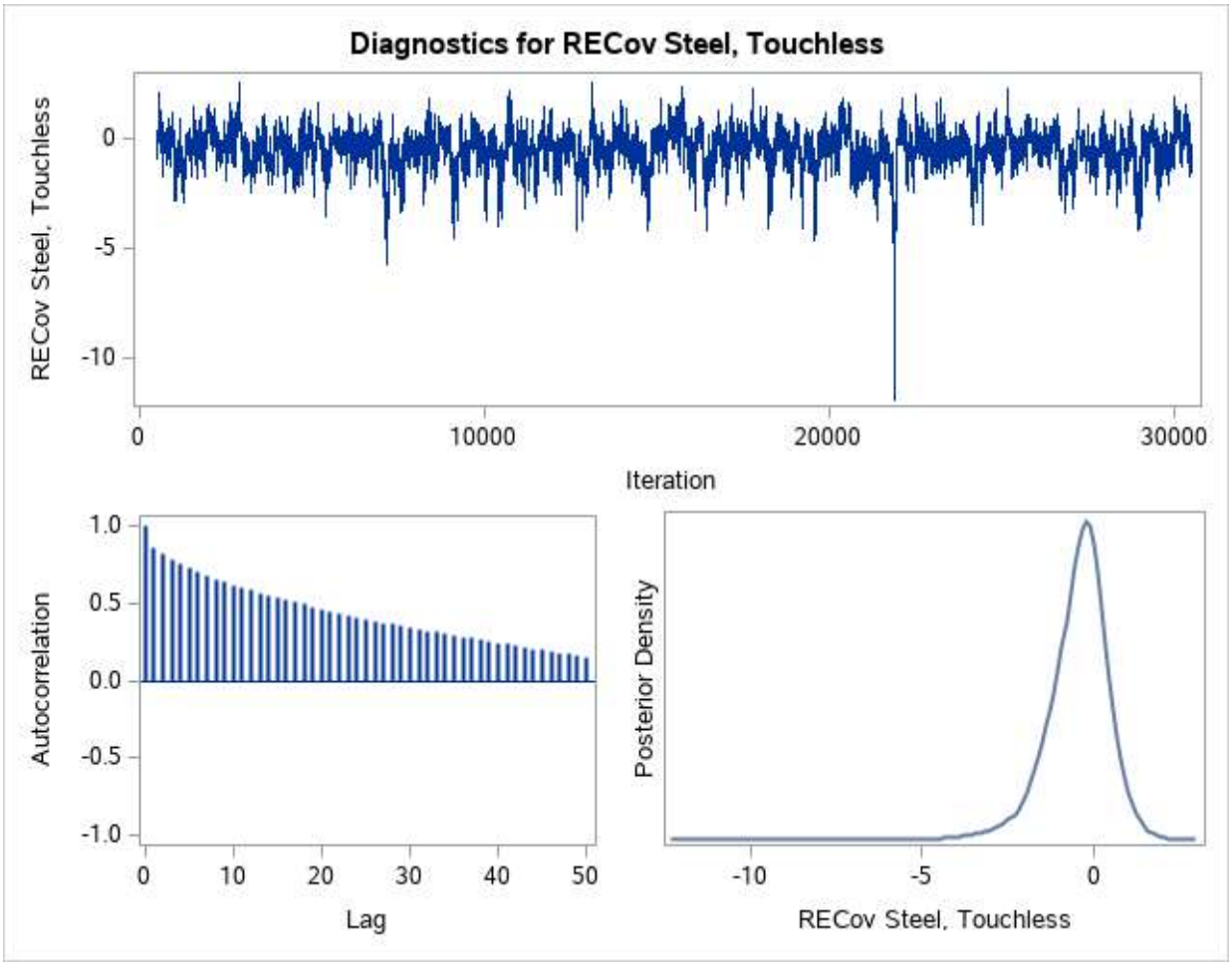


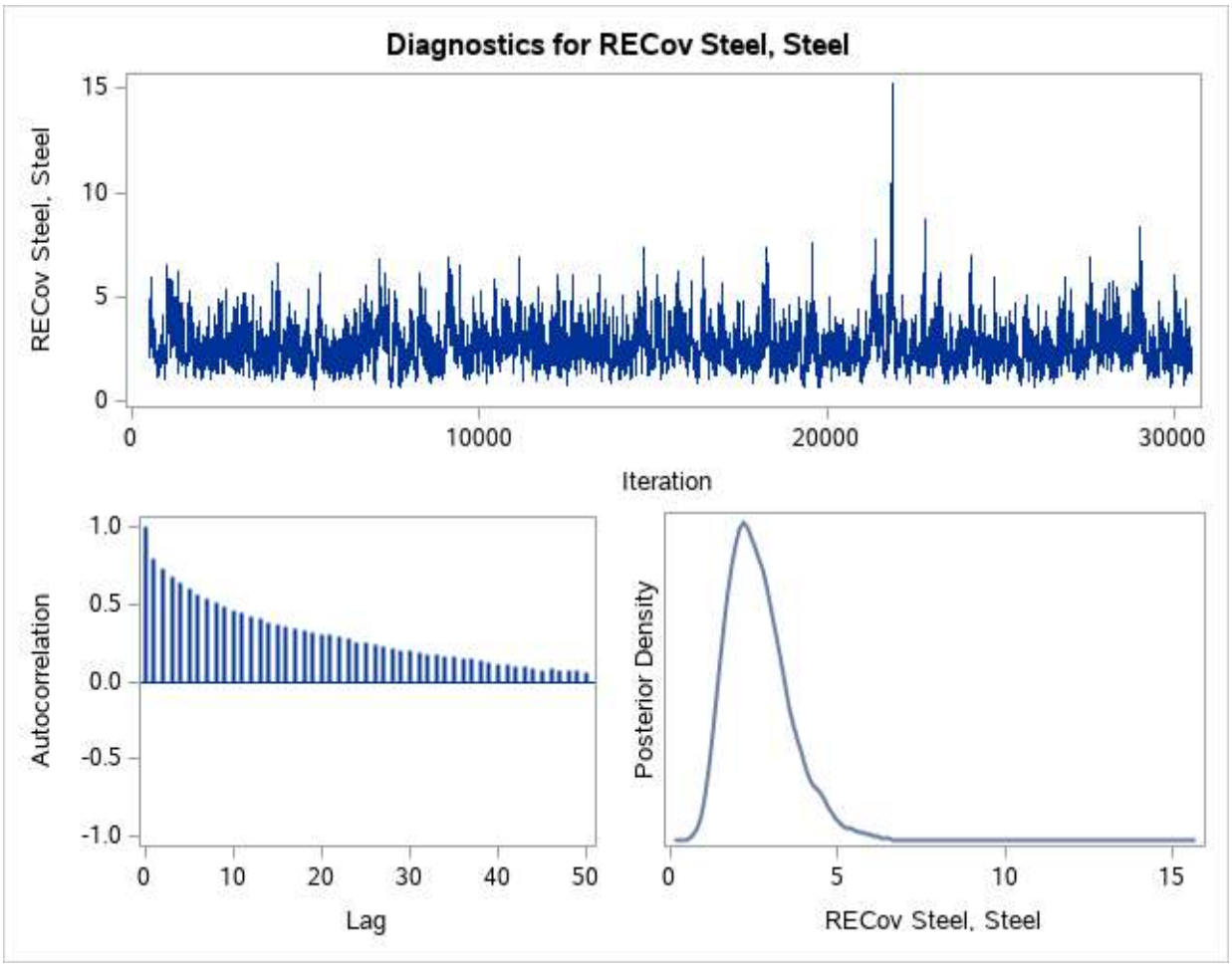

Diagnostics for Steel
Steel
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1.5
1.0
0.5
0.0
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
0.0
0.5
1.0
1.5
2.0
Steel

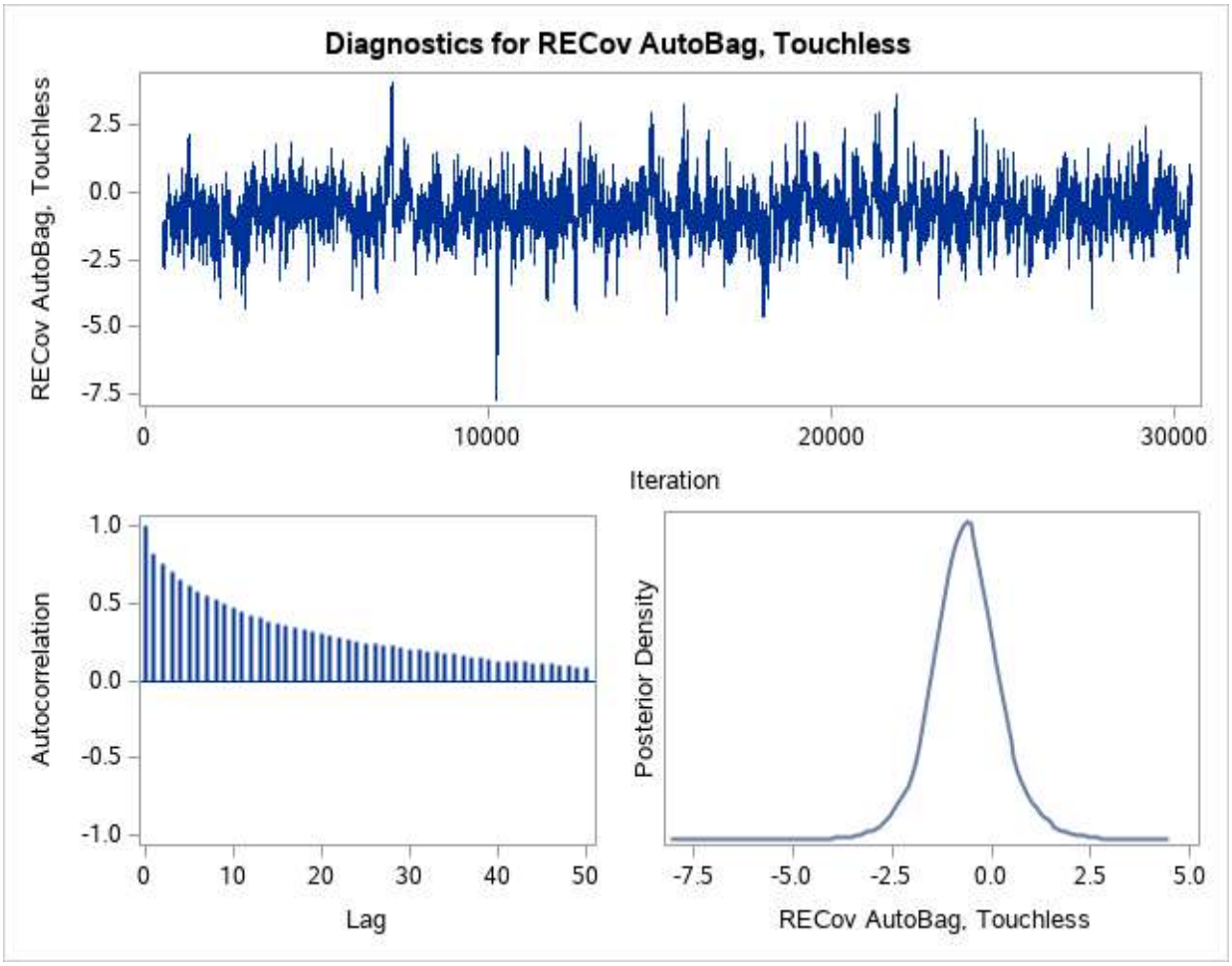

Diagnostics for AutoBag
AutoBag
4
3
2
1
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
1
2
3
4
AutoBag

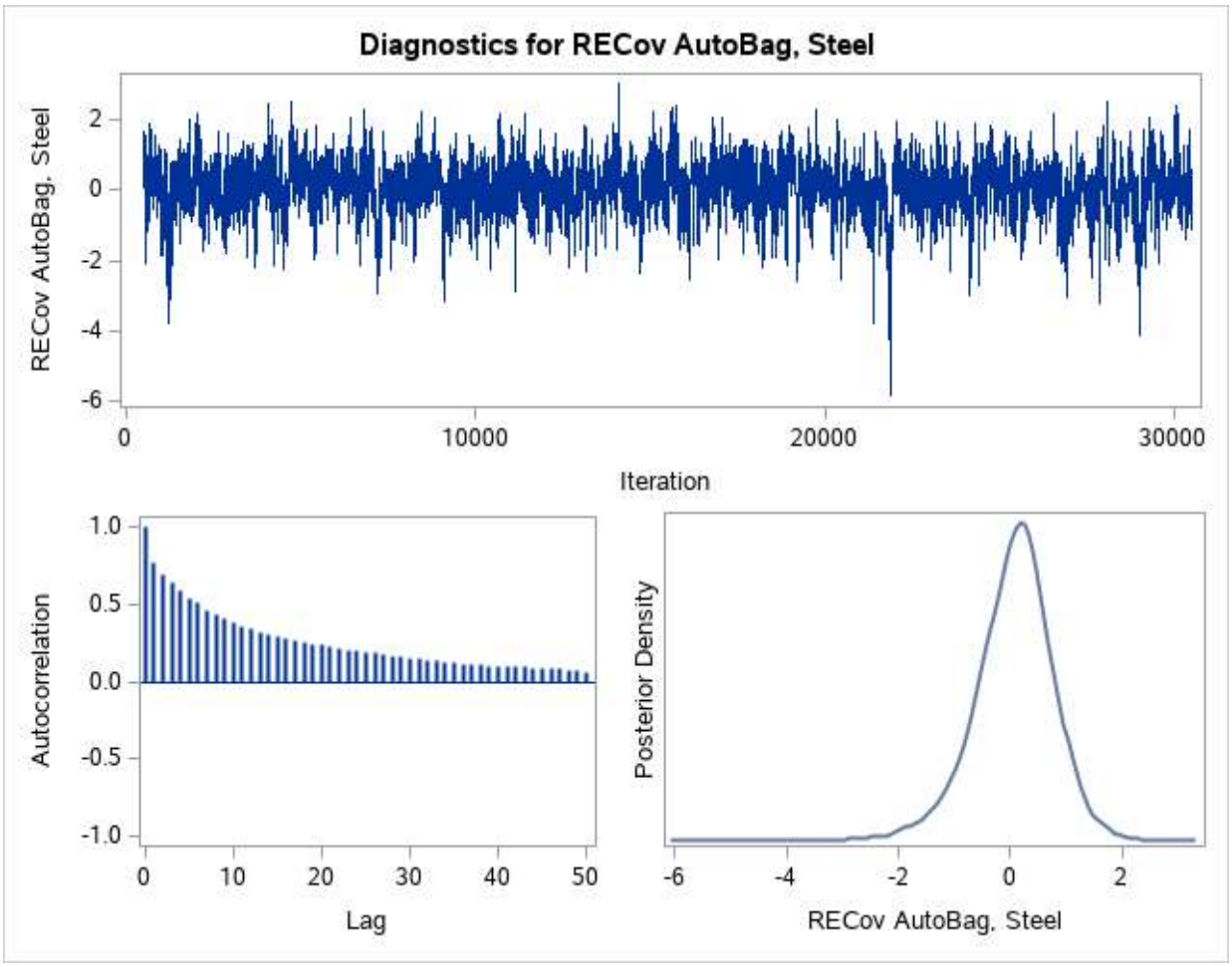

Diagnostics for Price80
Price80
-4
-6
-8
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-8
-6
-4
-2
Price80

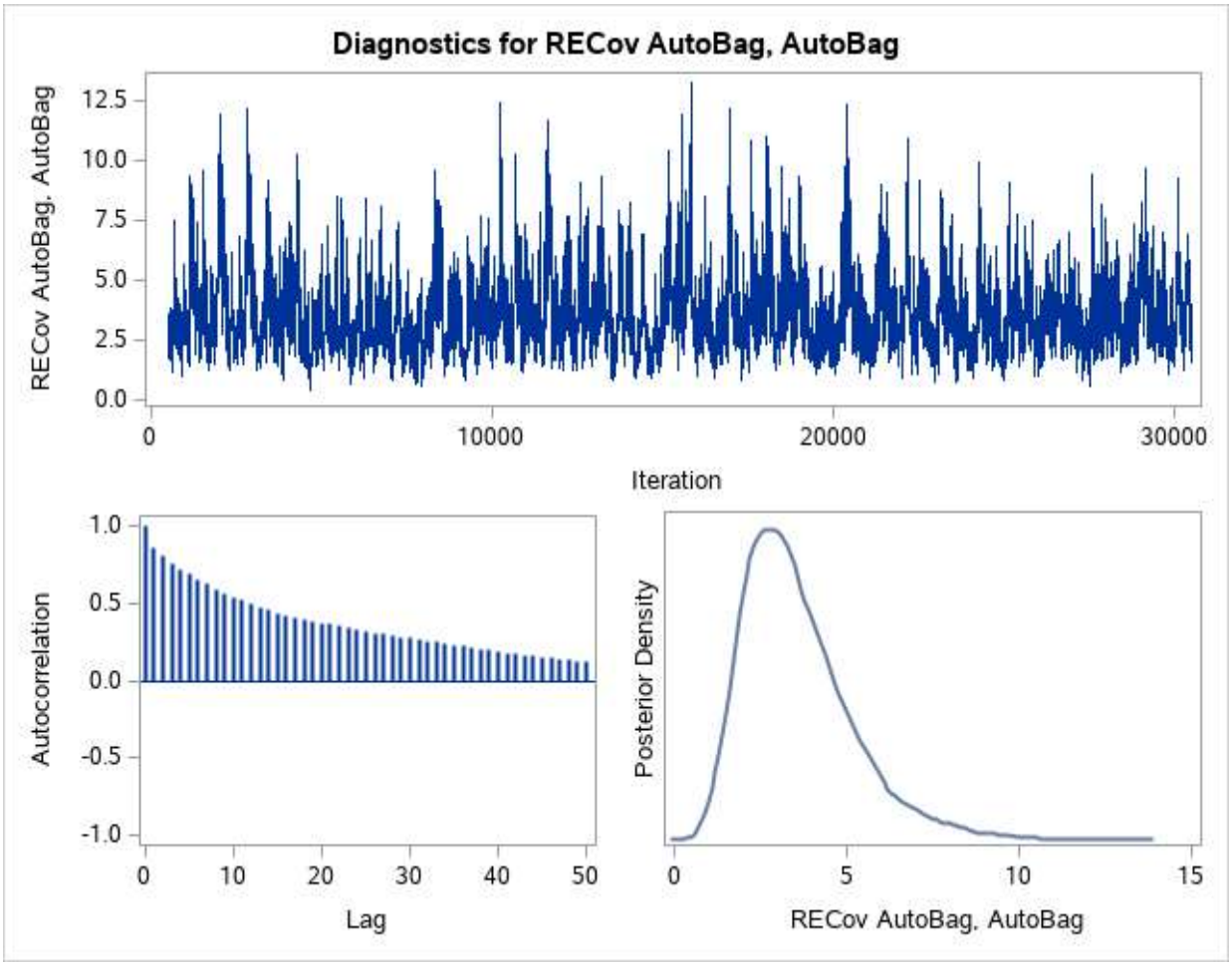

Diagnostics for RECov Touchless, Touchless
RECov Touchless, Touchl...
12.5
10.0
7.5
5.0
2.5
0.0
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
0.0
2.5
5.0
7.5
10.0
12.5
RECov Touchless, Touchless

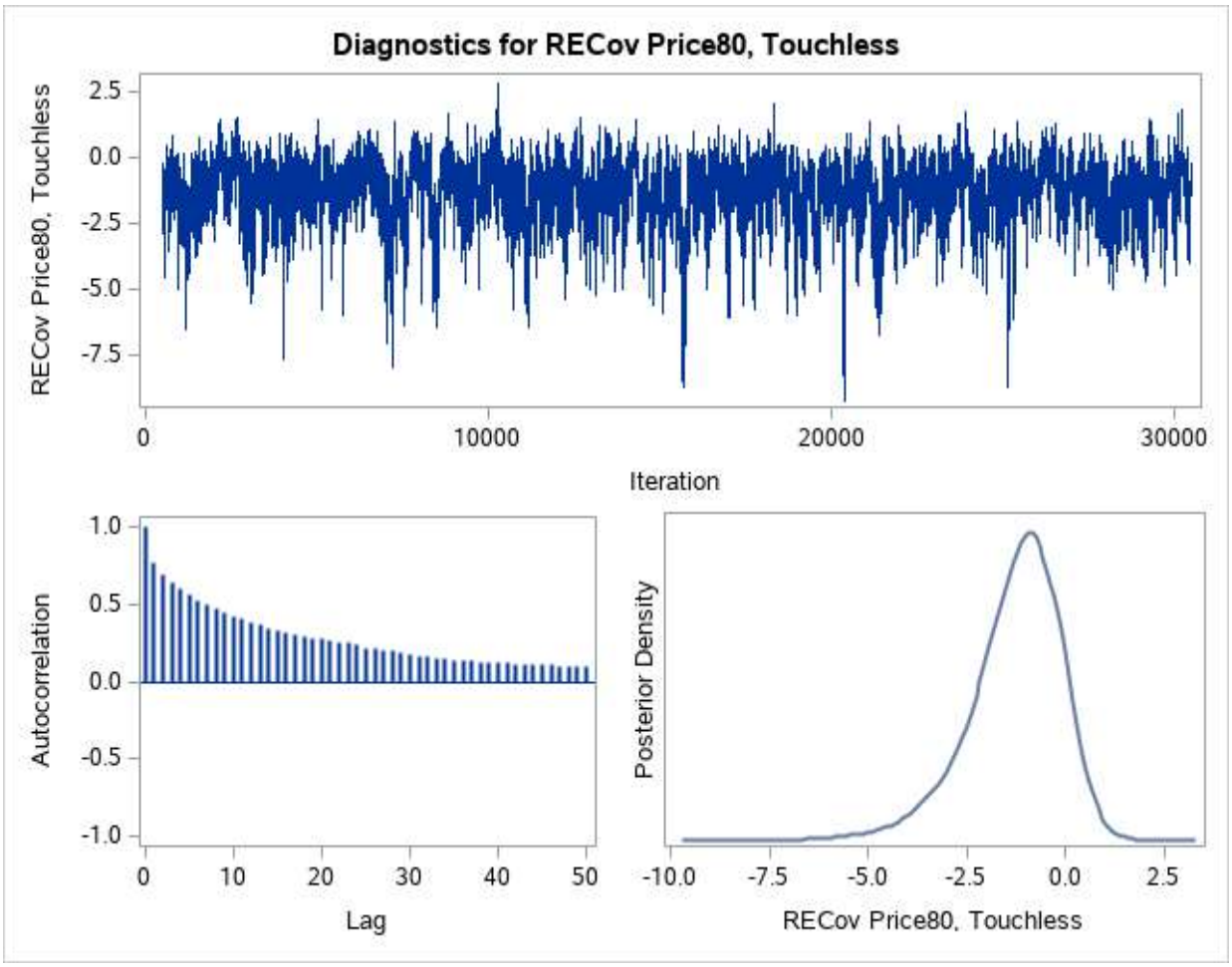

Diagnostics for RECov Steel, Touchless
RECov Steel, Touchless
0
-5
-10
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-10
-5
0
RECov Steel, Touchless

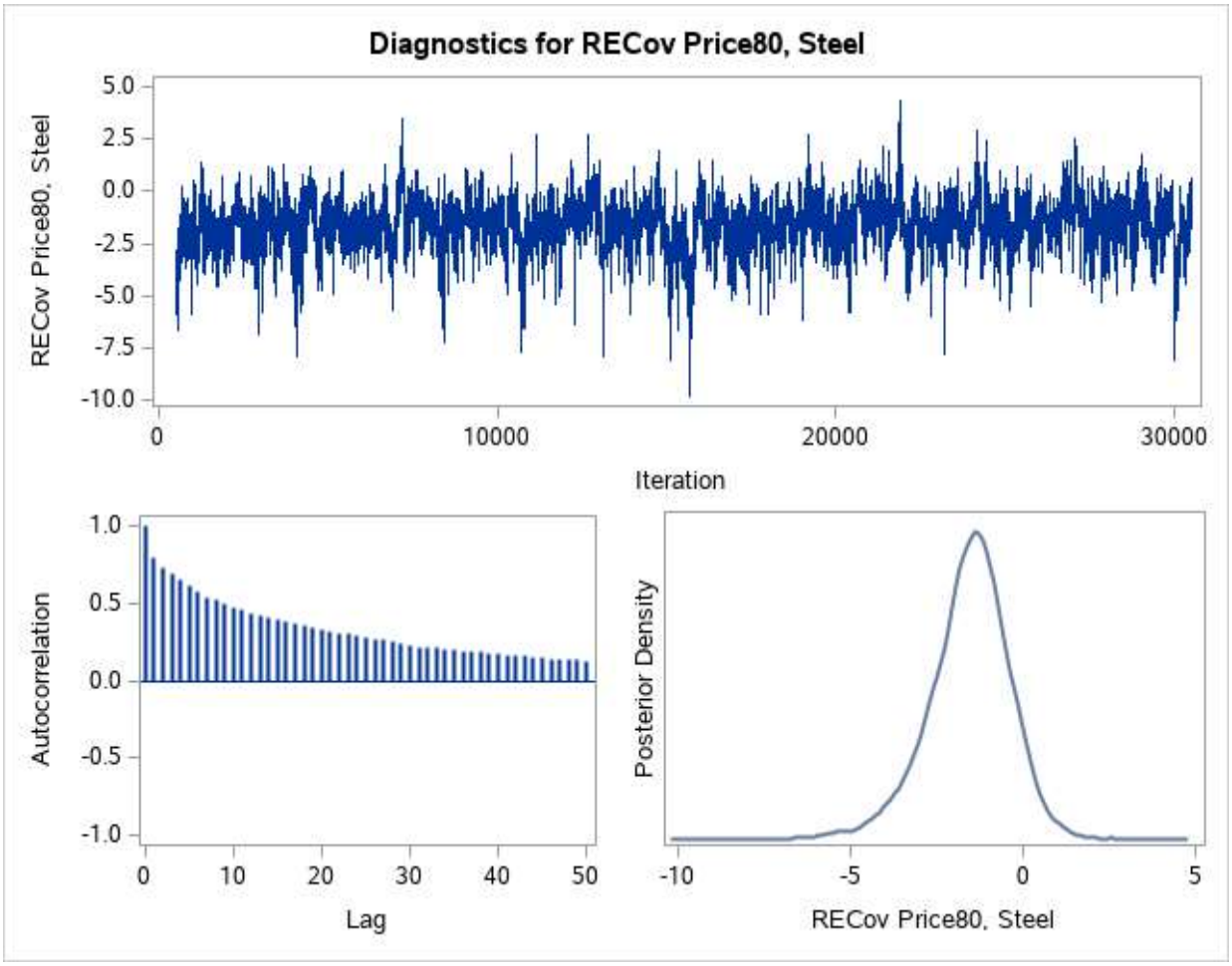

Diagnostics for RECov Steel, Steel
RECov Steel, Steel
15
10
5
0
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
0
5
10
15
RECov Steel, Steel

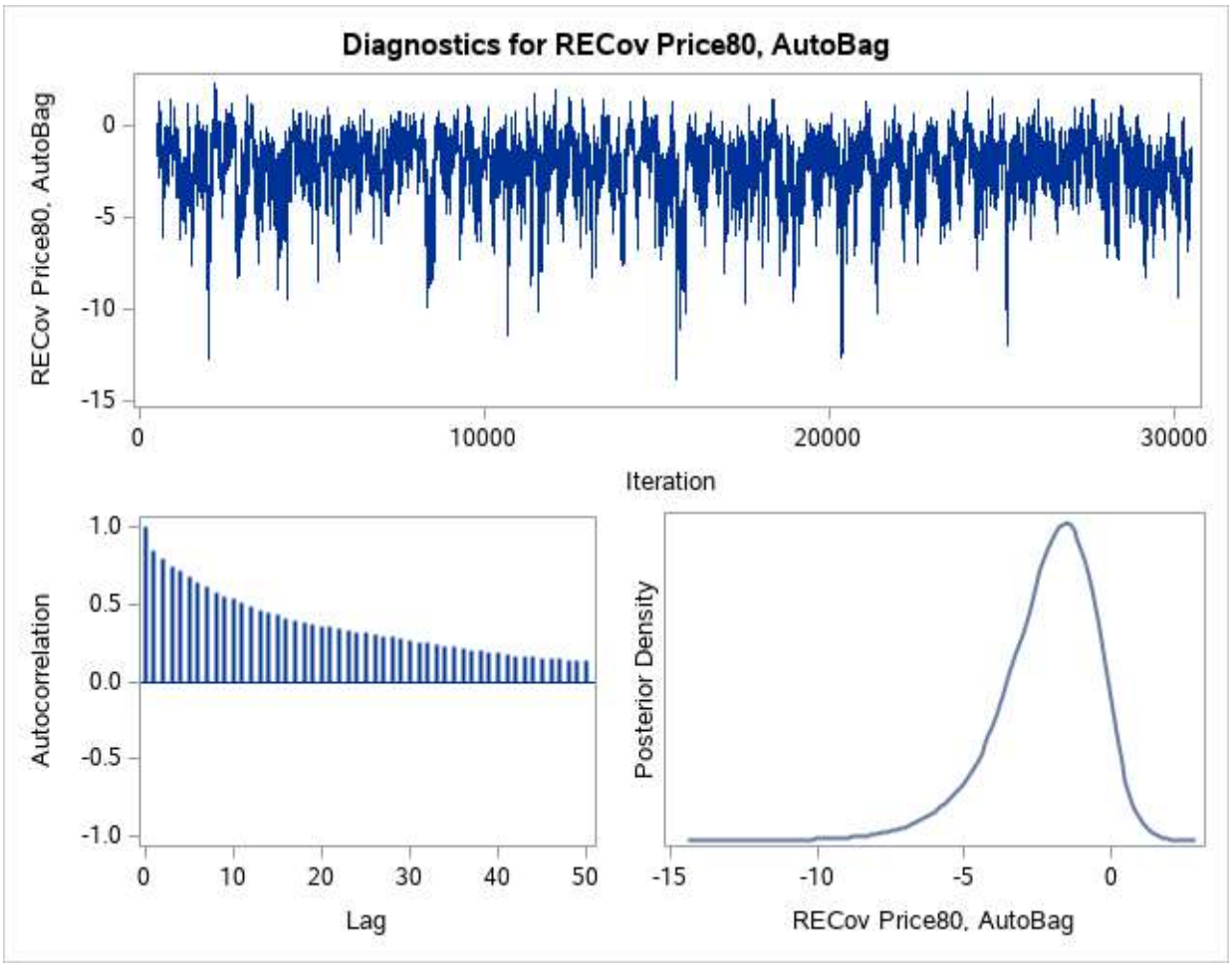

Diagnostics for RECov AutoBag, Touchless
RECov AutoBag, Touchless
2.5
0.0
-2.5
-5.0
-7.5
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-7.5
-5.0
-2.5
0.0
2.5
5.0
RECov AutoBag, Touchless

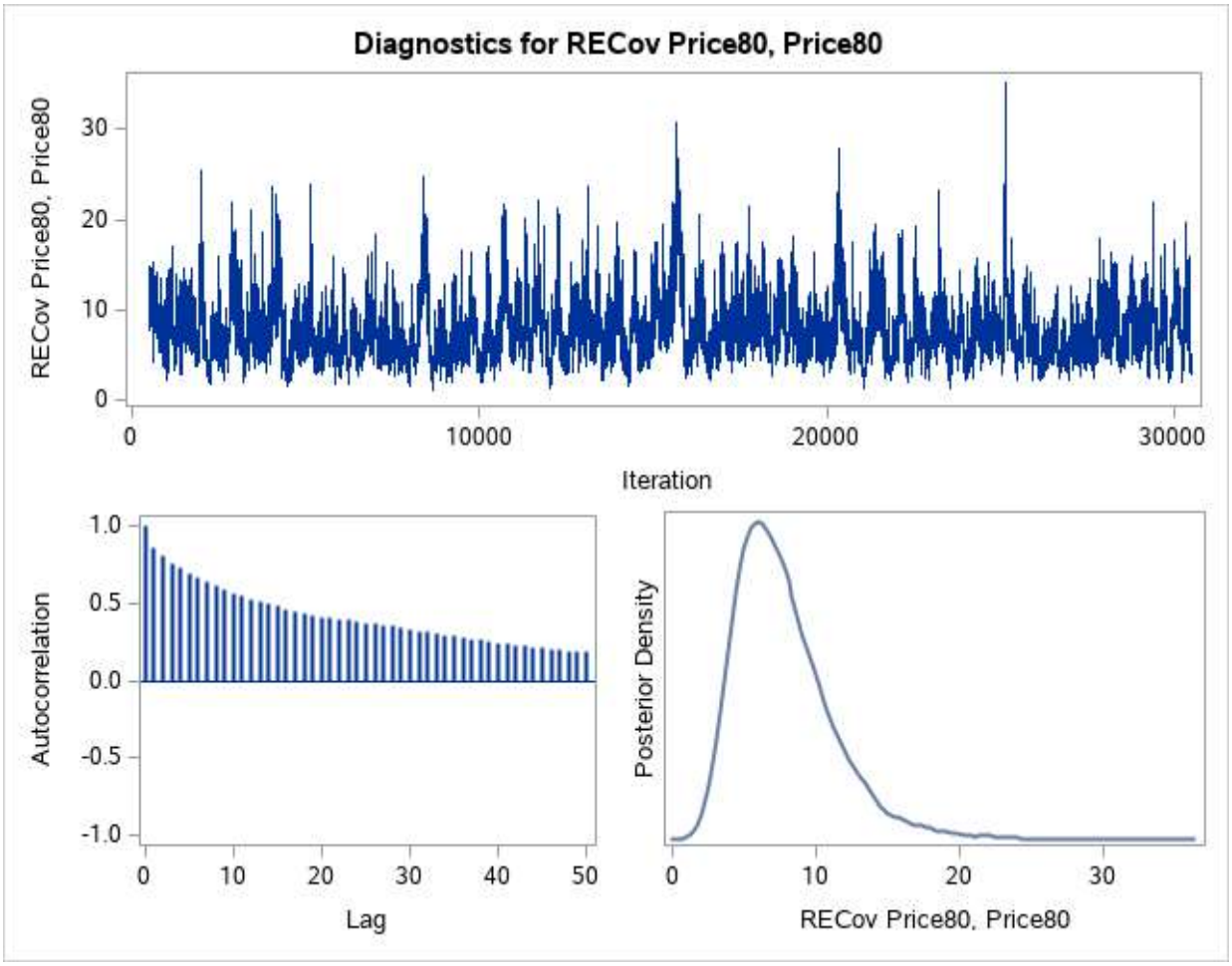

Diagnostics for RECov AutoBag, Steel
RECov AutoBag, Steel
2
0
-2
-4
-6
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-6
-4
-2
0
2
RECov AutoBag, Steel

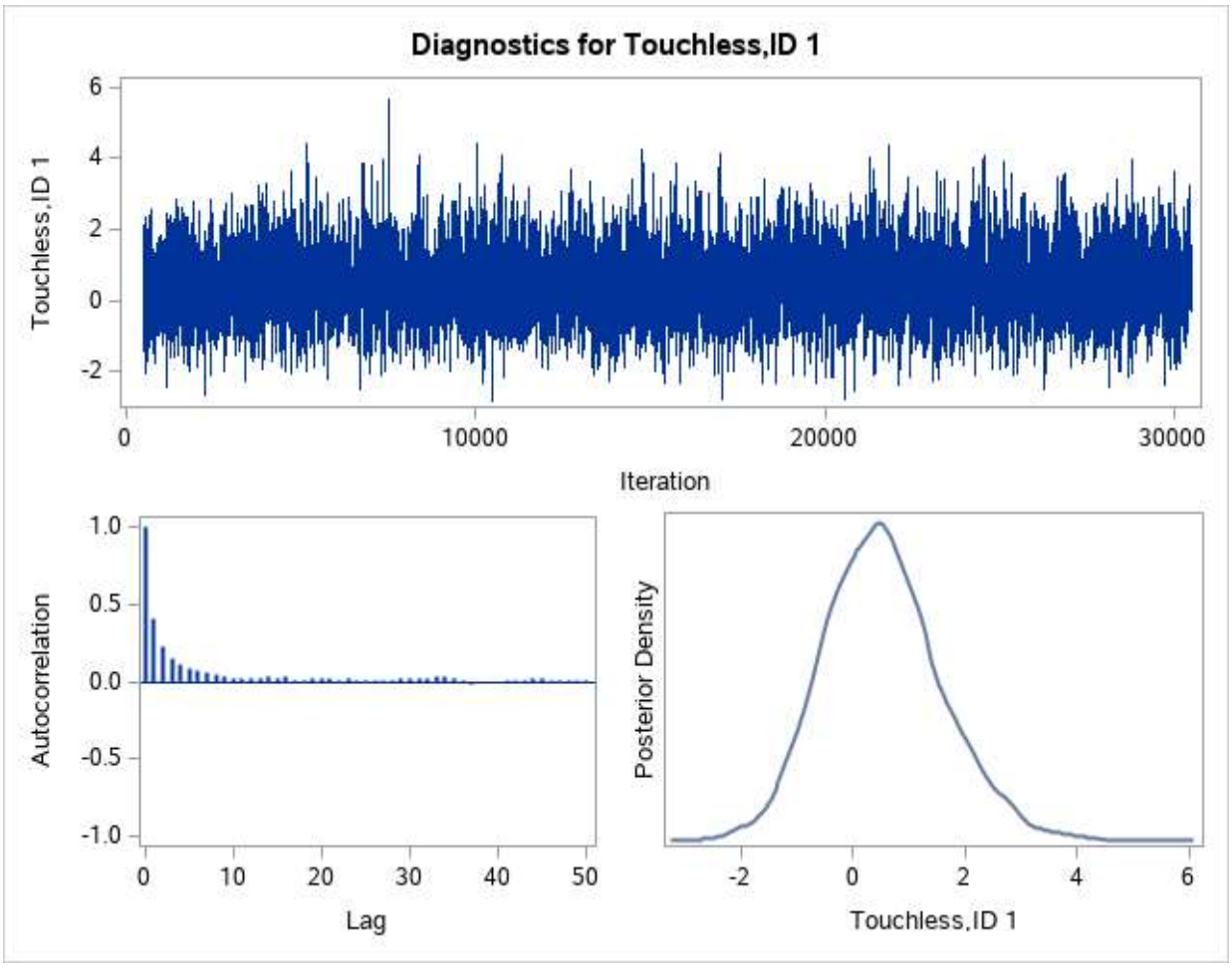

Diagnostics for RECov AutoBag, AutoBag
RECov AutoBag, AutoBag
12.5
10.0
7.5
5.0
2.5
0.0
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
0
5
10
15
RECov AutoBag, AutoBag

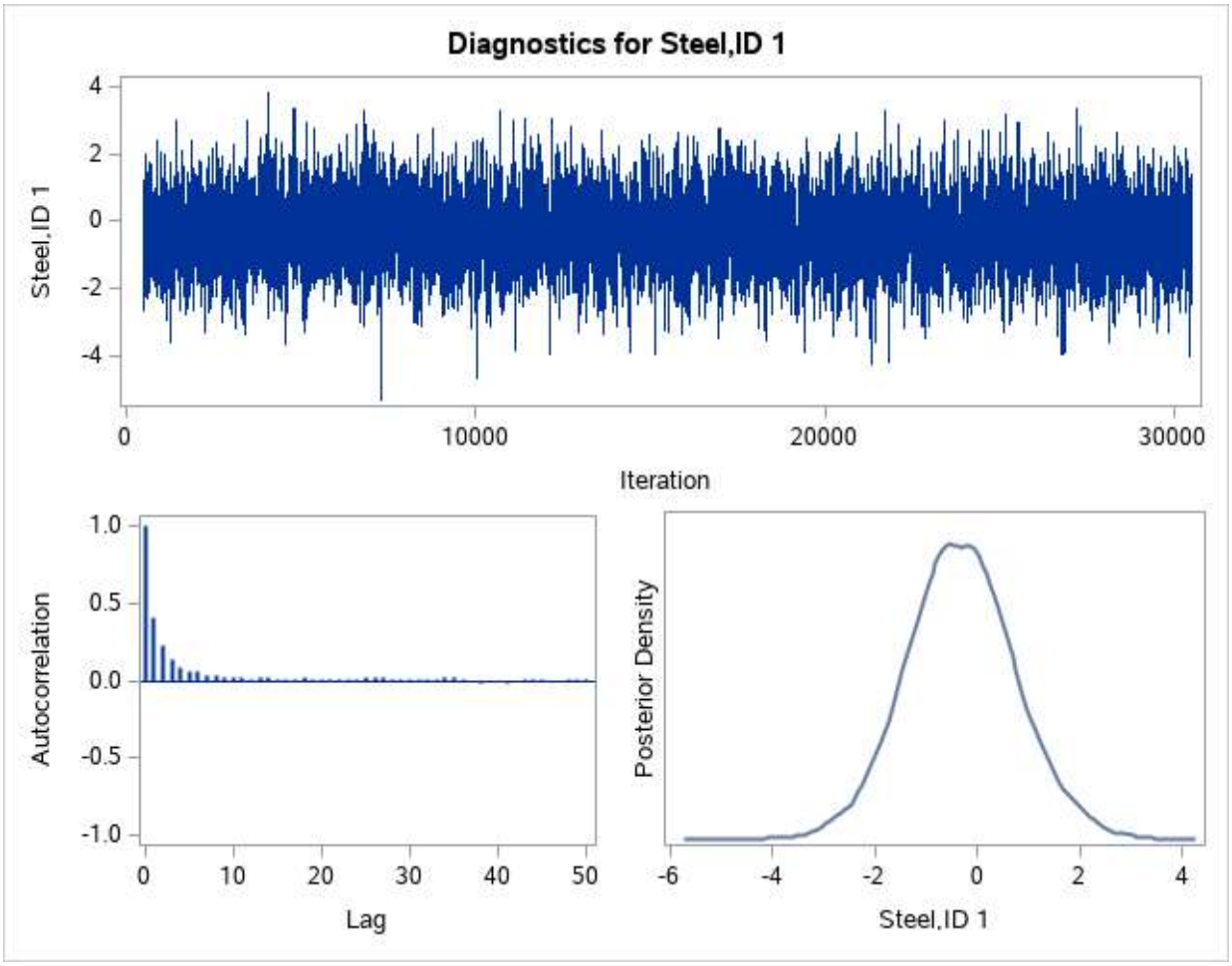

Diagnostics for RECov Price80, Touchless
RECov Price80, Touchless
2.5
0.0
-2.5
-5.0
-7.5
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-10.0
-7.5
-5.0
-2.5
0.0
2.5
RECov Price80, Touchless

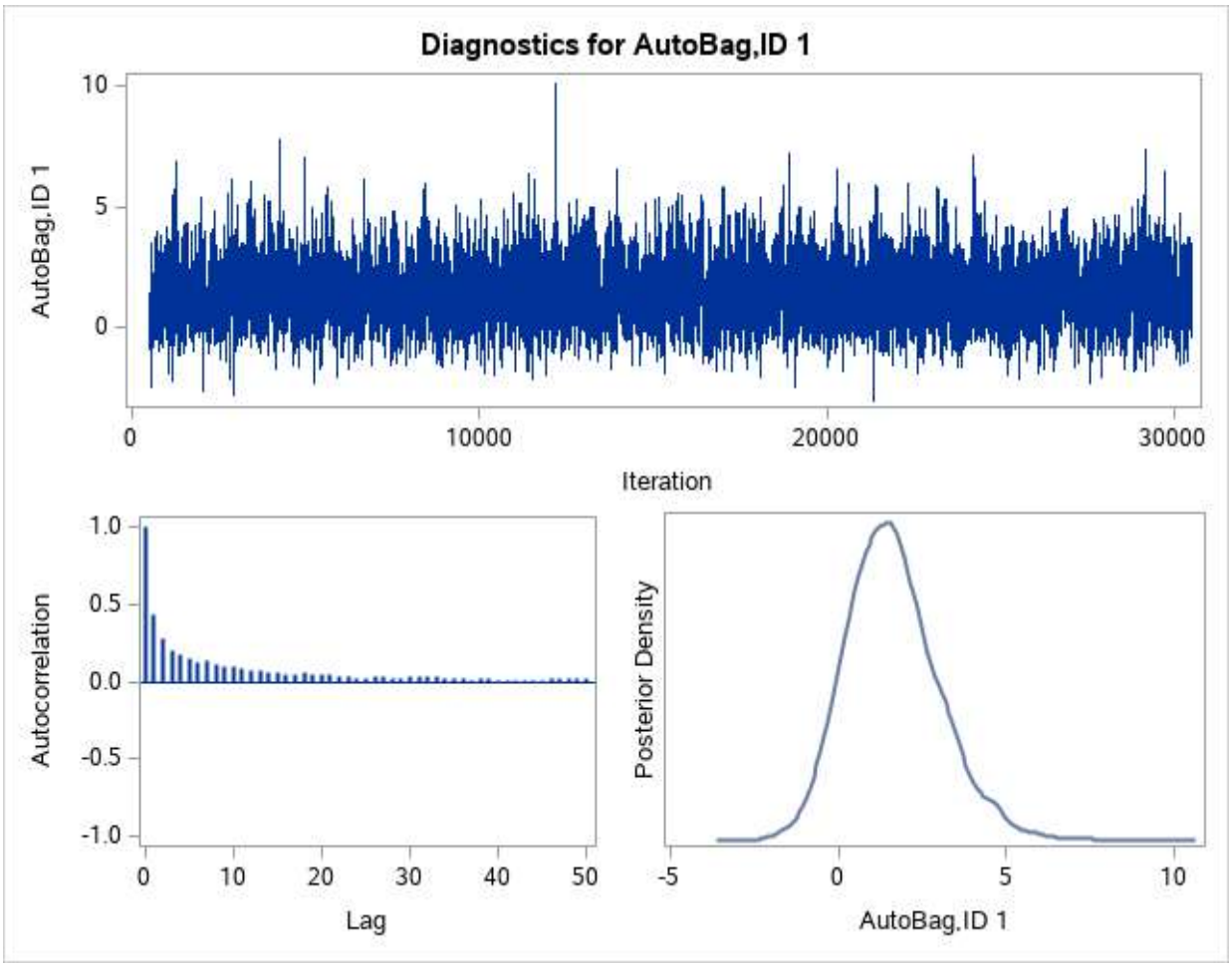

Diagnostics for RECov Price80, Steel
RECov Price80, Steel
5.0
2.5
0.0
-2.5
-5.0
-7.5
-10.0
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-10
-5
0
5
RECov Price80, Steel

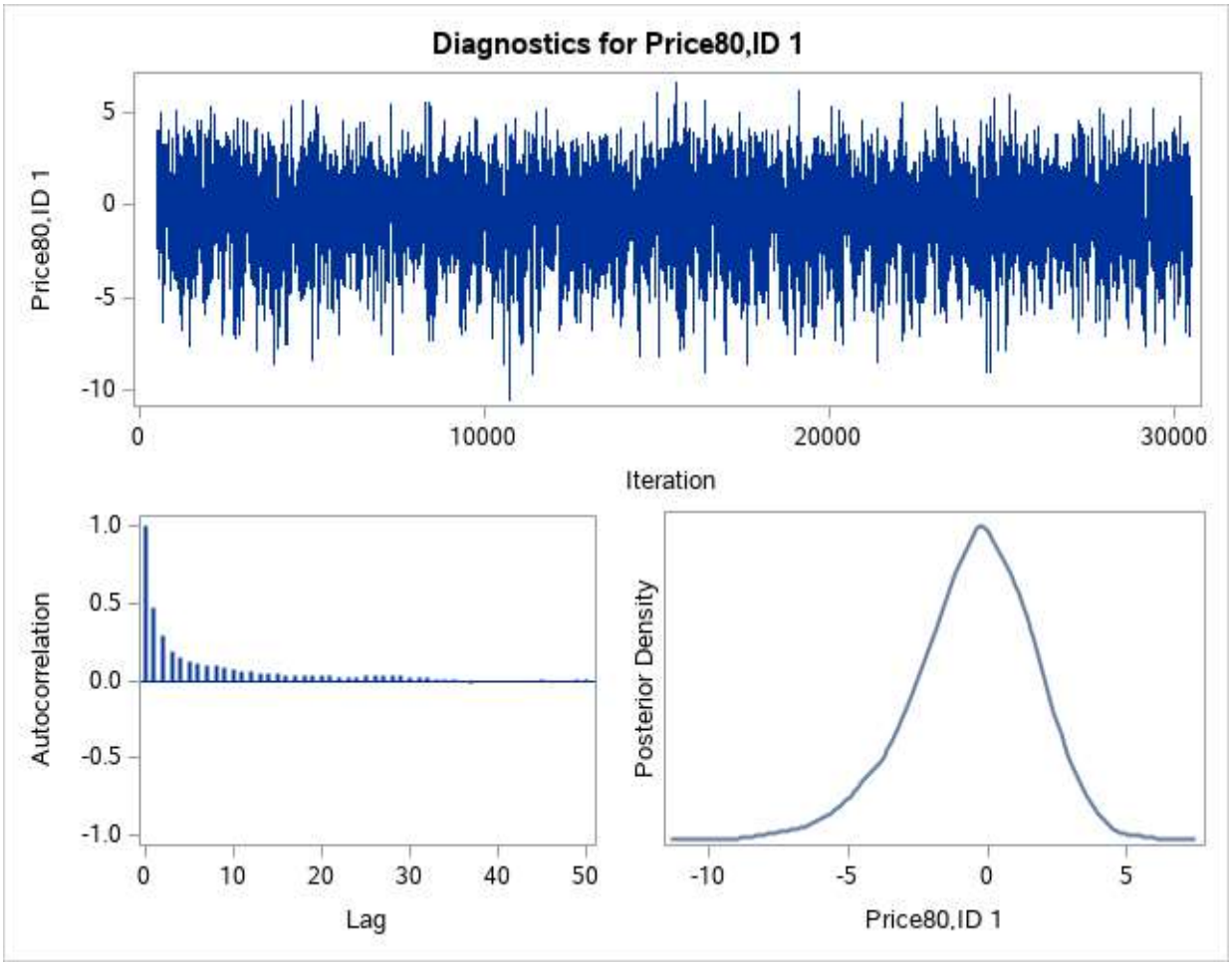

Diagnostics for RECov Price80, AutoBag
RECov Price80, AutoBag
0
-5
-10
-15
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-15
-10
-5
0
RECov Price80, AutoBag

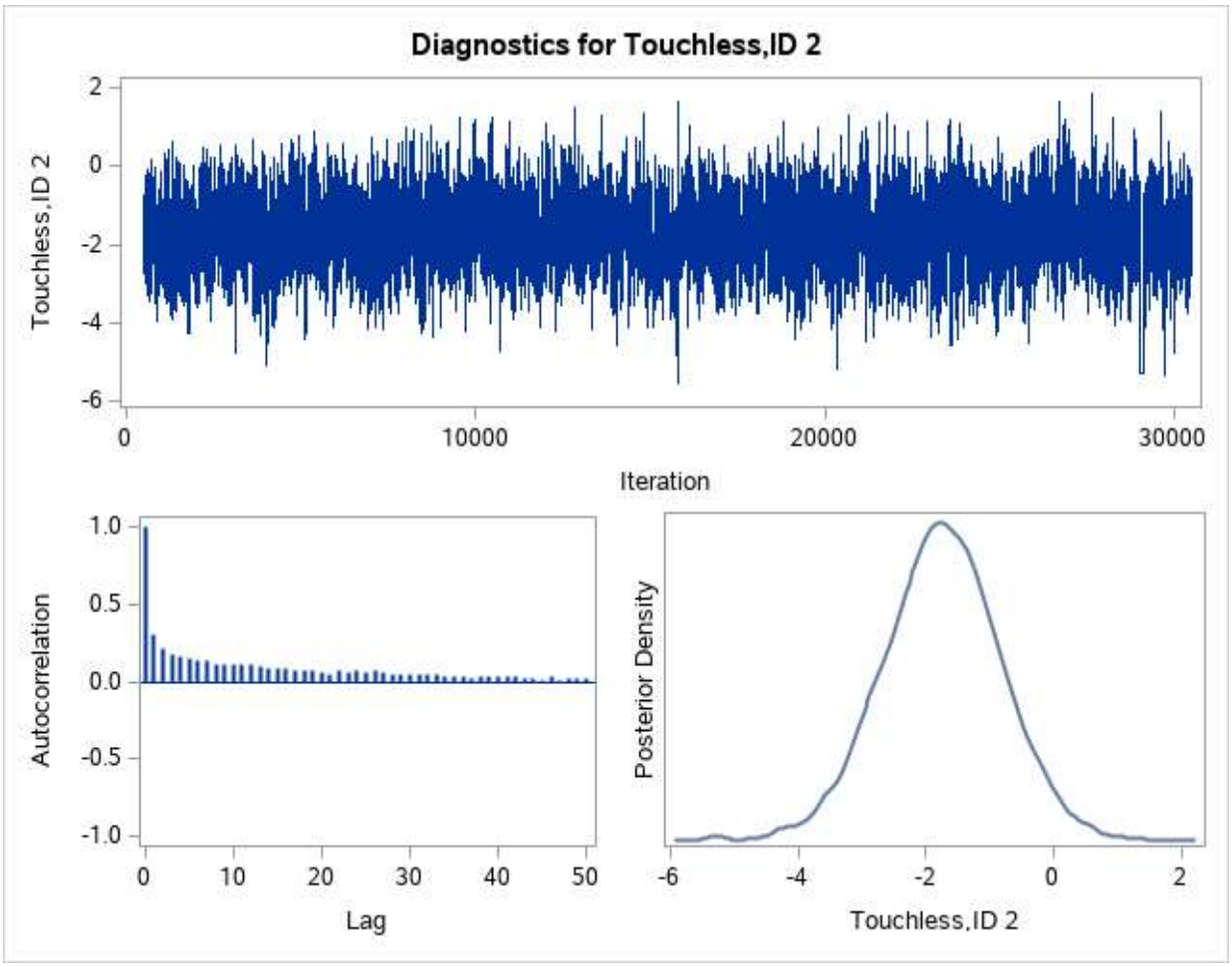

Diagnostics for RECov Price80, Price80
RECov Price80, Price80
30
20
10
0
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
0
10
20
30
RECov Price80, Price80

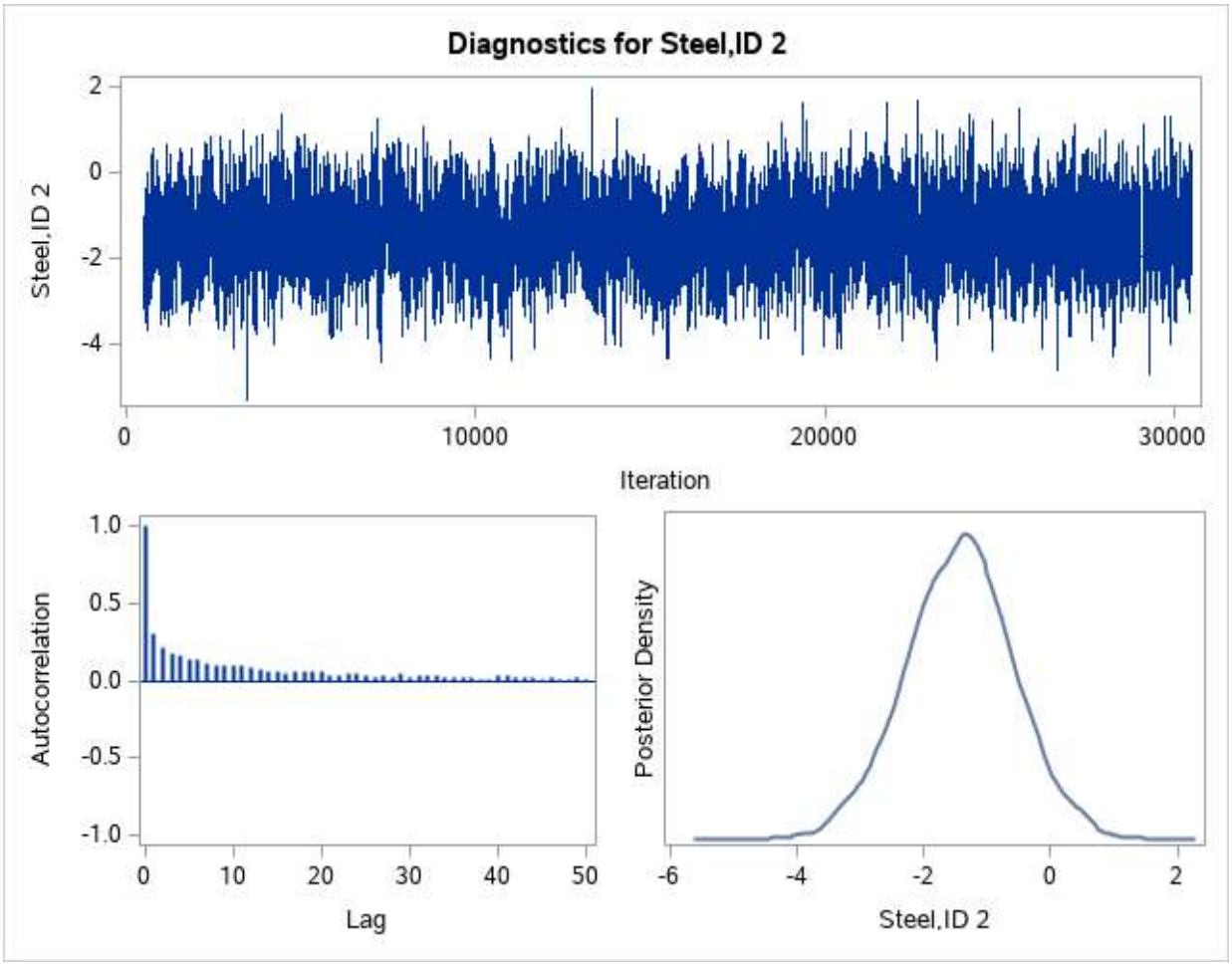

Diagnostics for Touchless,ID 1
Touchless,ID 1
6
4
2
0
-2
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-2
0
2
4
6
Touchless,ID 1

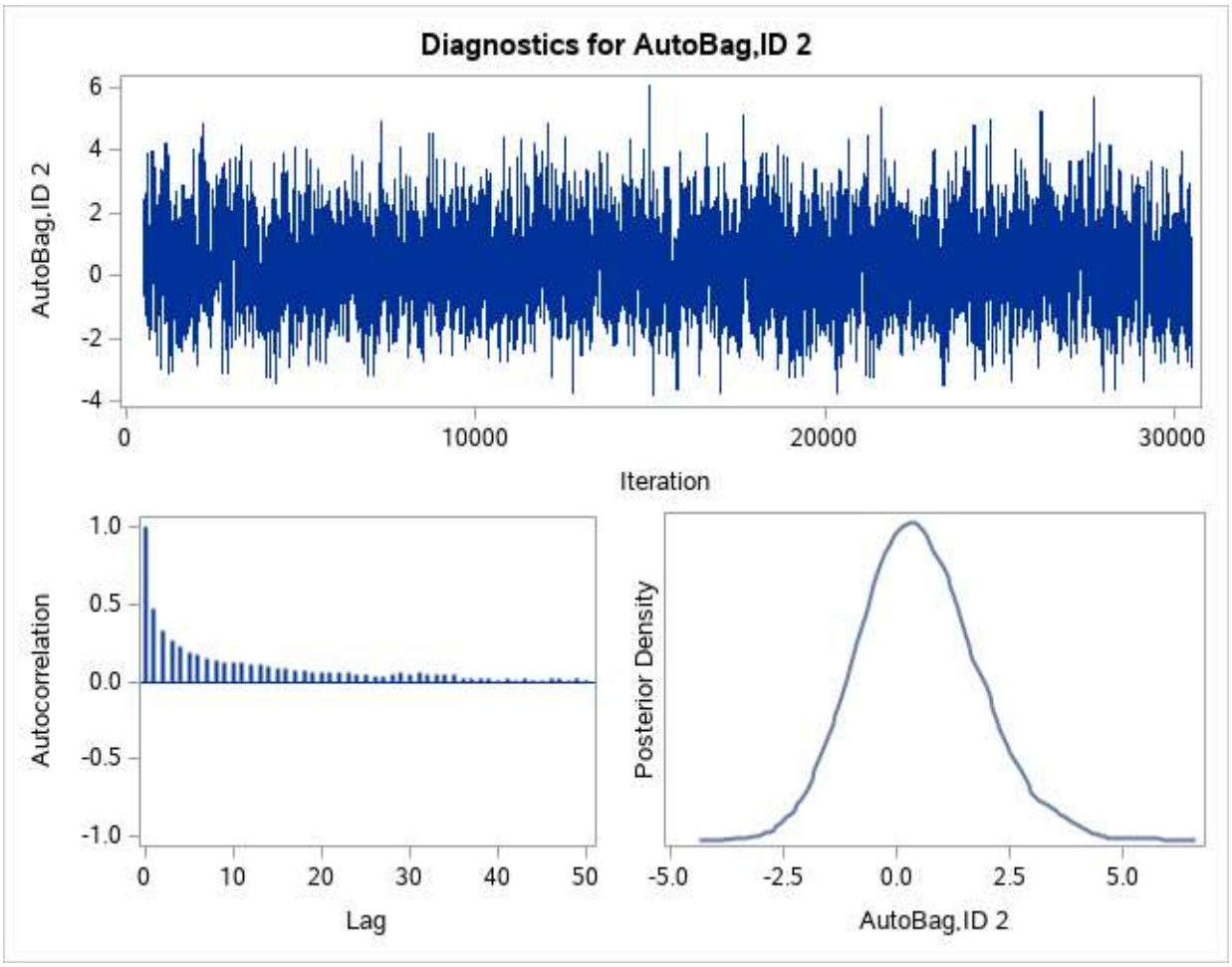

Diagnostics for Steel,ID 1
Steel,ID 1
4
2
0
-2
-4
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-6
-4
-2
0
2
4
Steel,ID 1

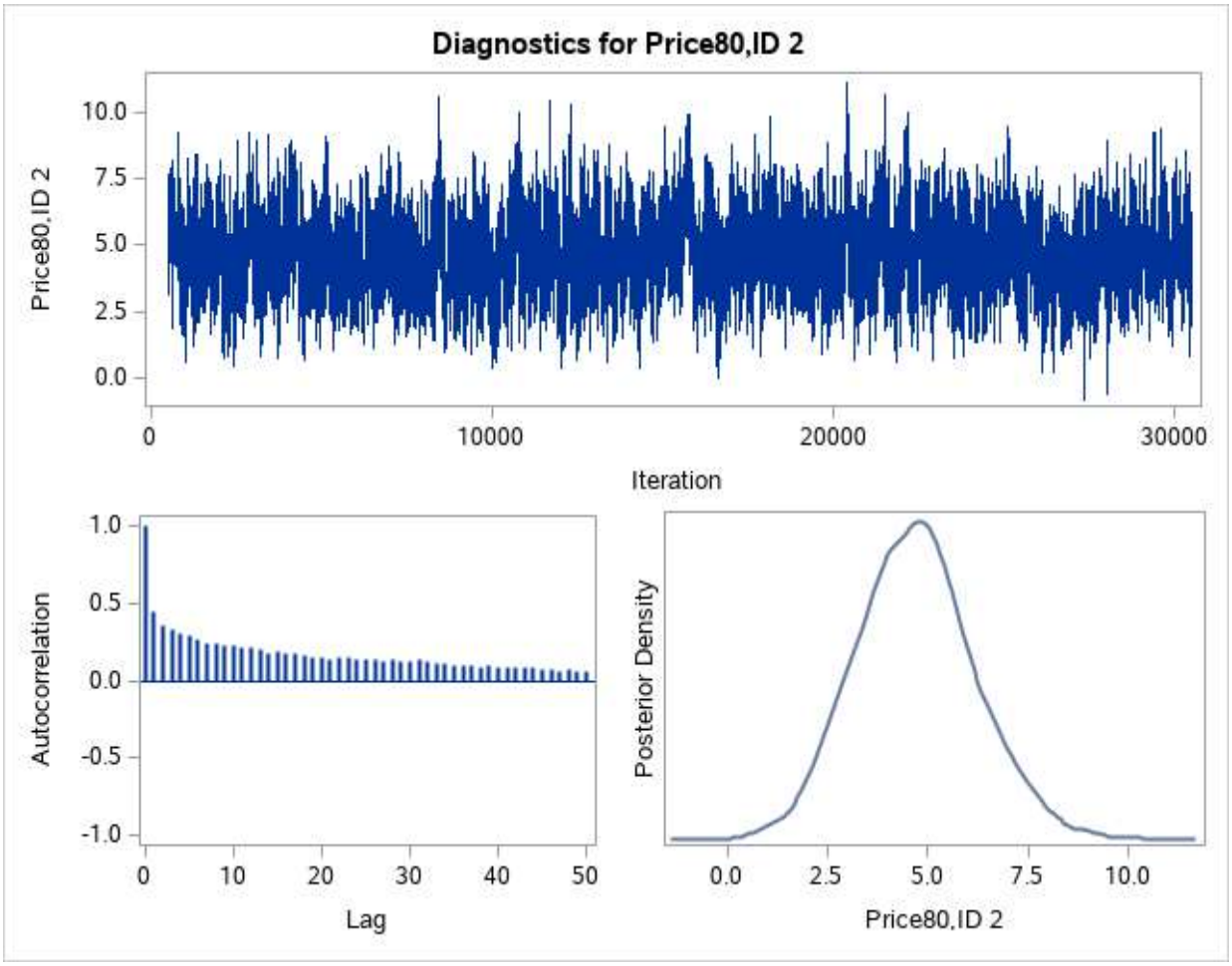

Diagnostics for AutoBag,ID 1
AutoBag,ID 1
10
5
0
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-5
0
5
10
AutoBag,ID 1

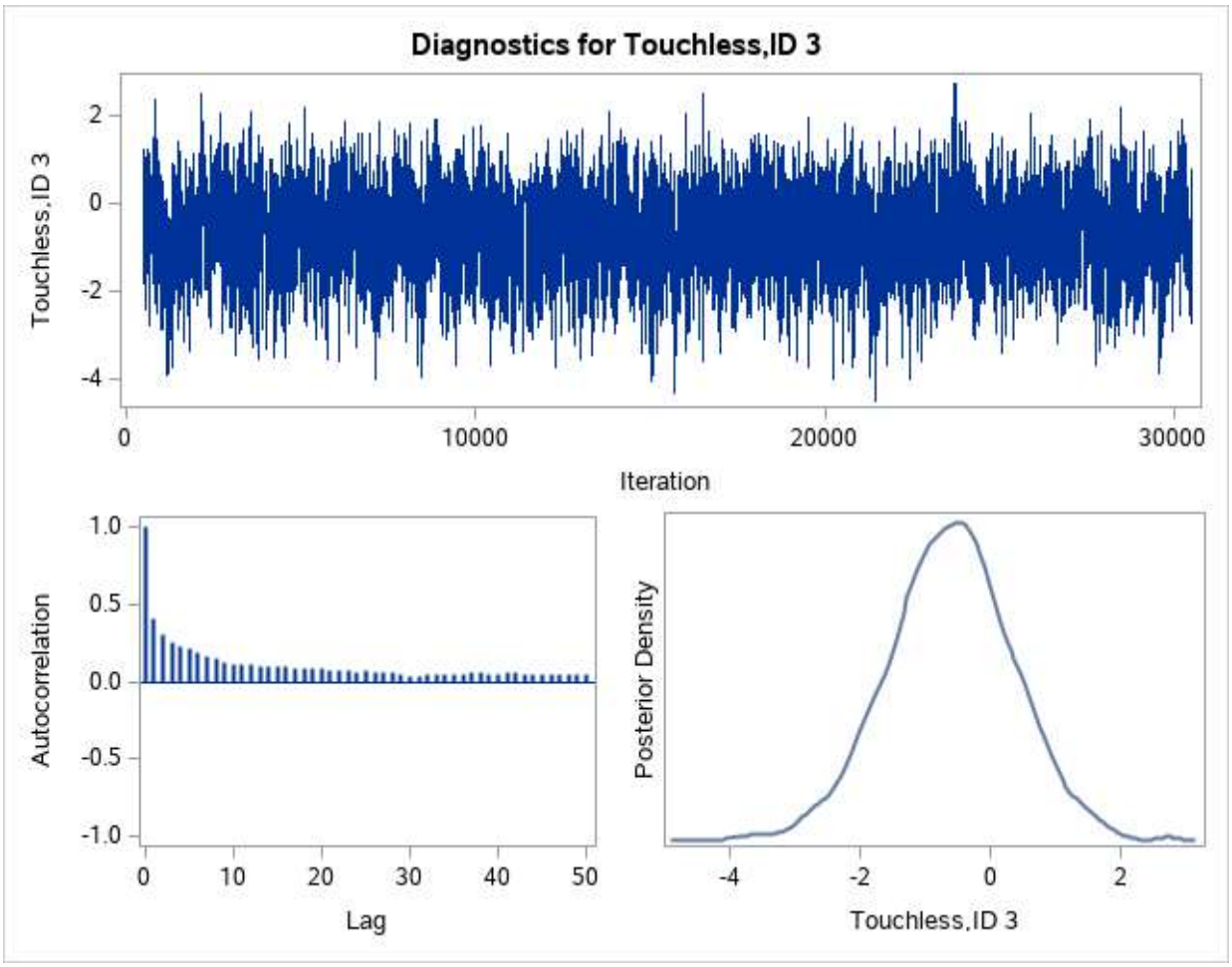

Diagnostics for Price80,ID 1
Price80,ID 1
5
0
-5
-10
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-10
-5
0
5
Price80,ID 1

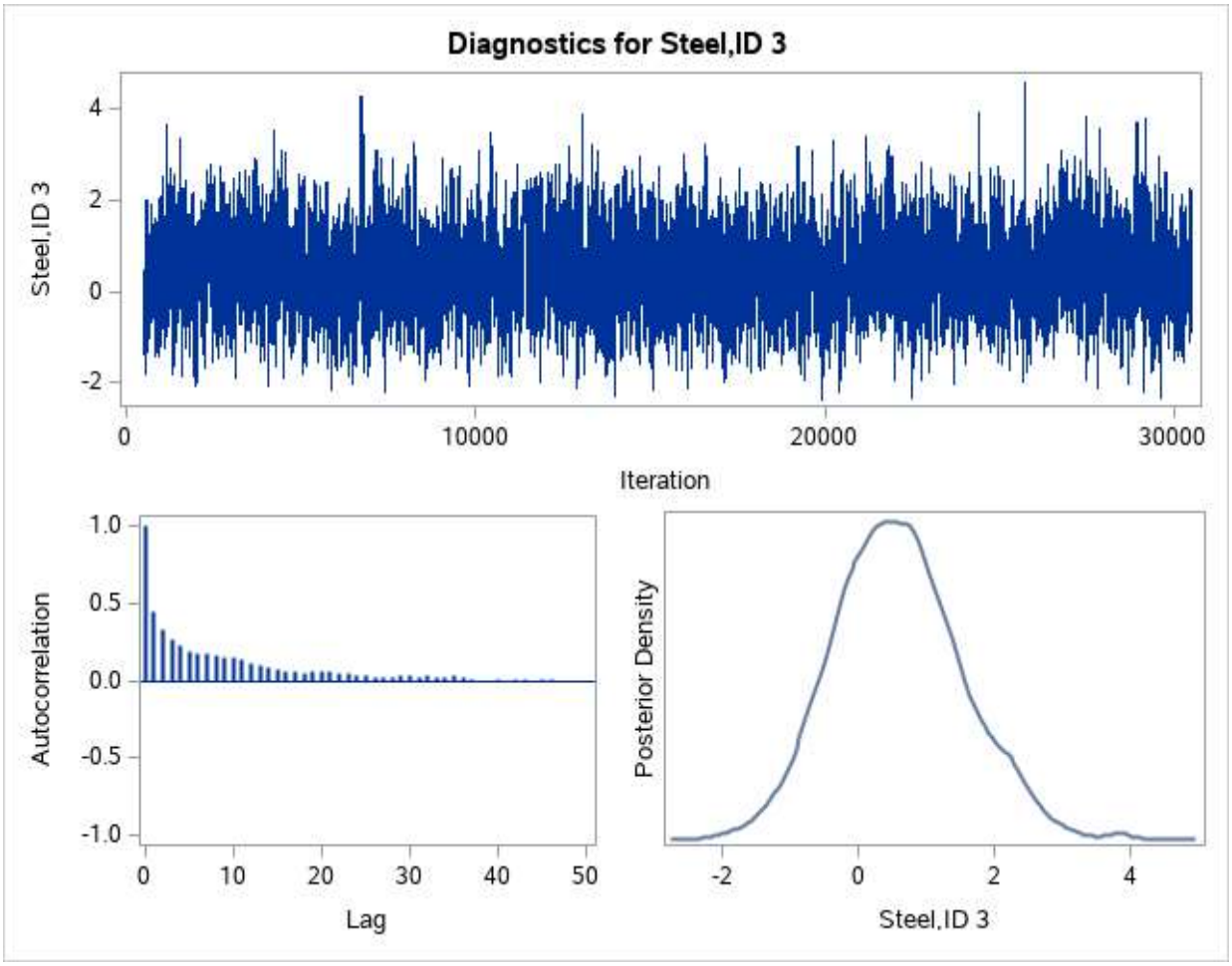

Diagnostics for Touchless,ID 2
Touchless,ID 2
2
0
-2
-4
-6
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-6
-4
-2
0
2
Touchless,ID 2

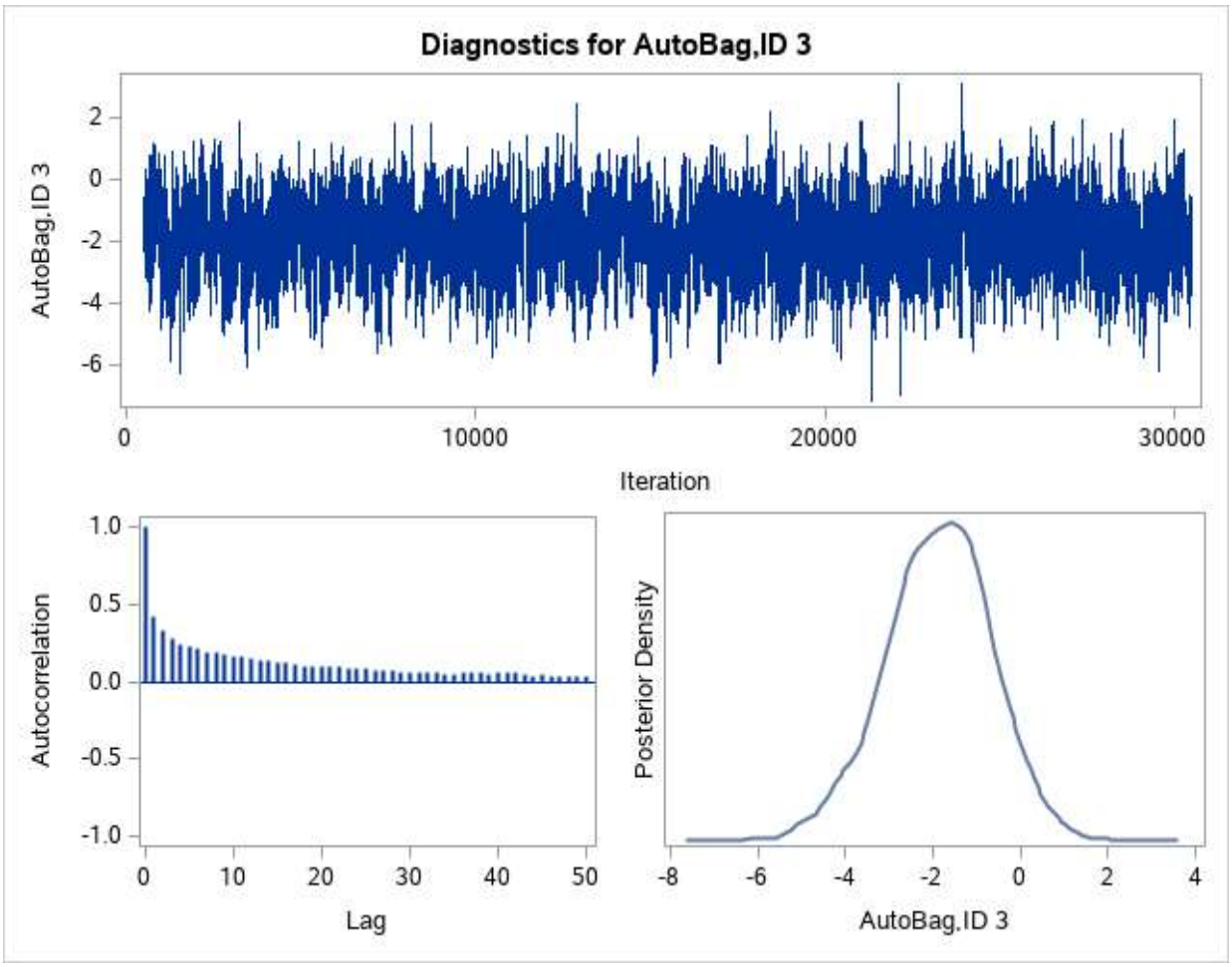

Diagnostics for Steel,ID 2
Steel,ID 2
2
0
-2
-4
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-6
-4
-2
0
2
Steel,ID 2

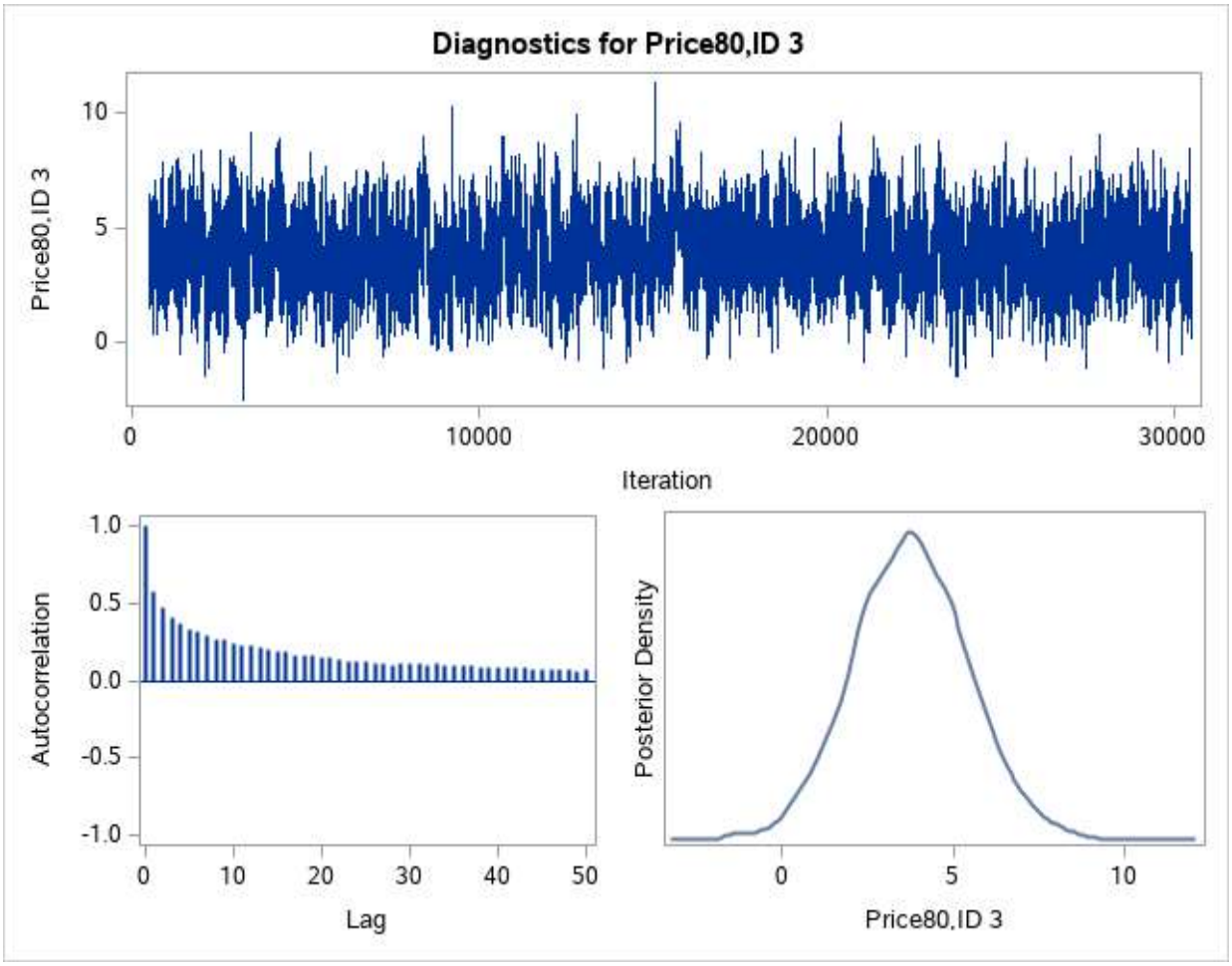

Diagnostics for AutoBag,ID 2
AutoBag,ID 2
6
4
2
0
-2
-4
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-5.0
-2.5
0.0
2.5
5.0
AutoBag,ID 2

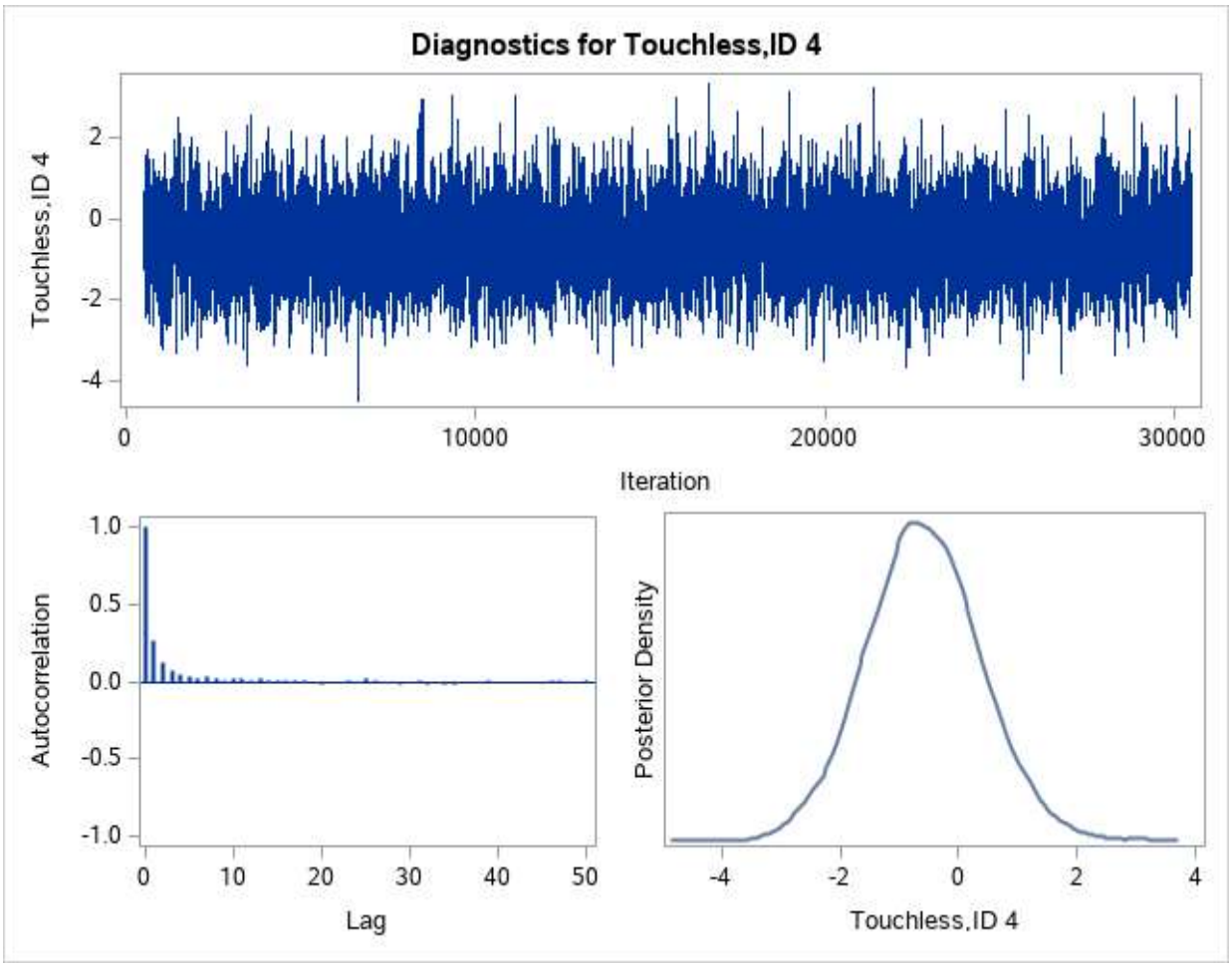

Diagnostics for Price80,ID 2
Price80,ID 2
10.0
7.5
5.0
2.5
0.0
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
0.0
2.5
5.0
7.5
10.0
Price80,ID 2

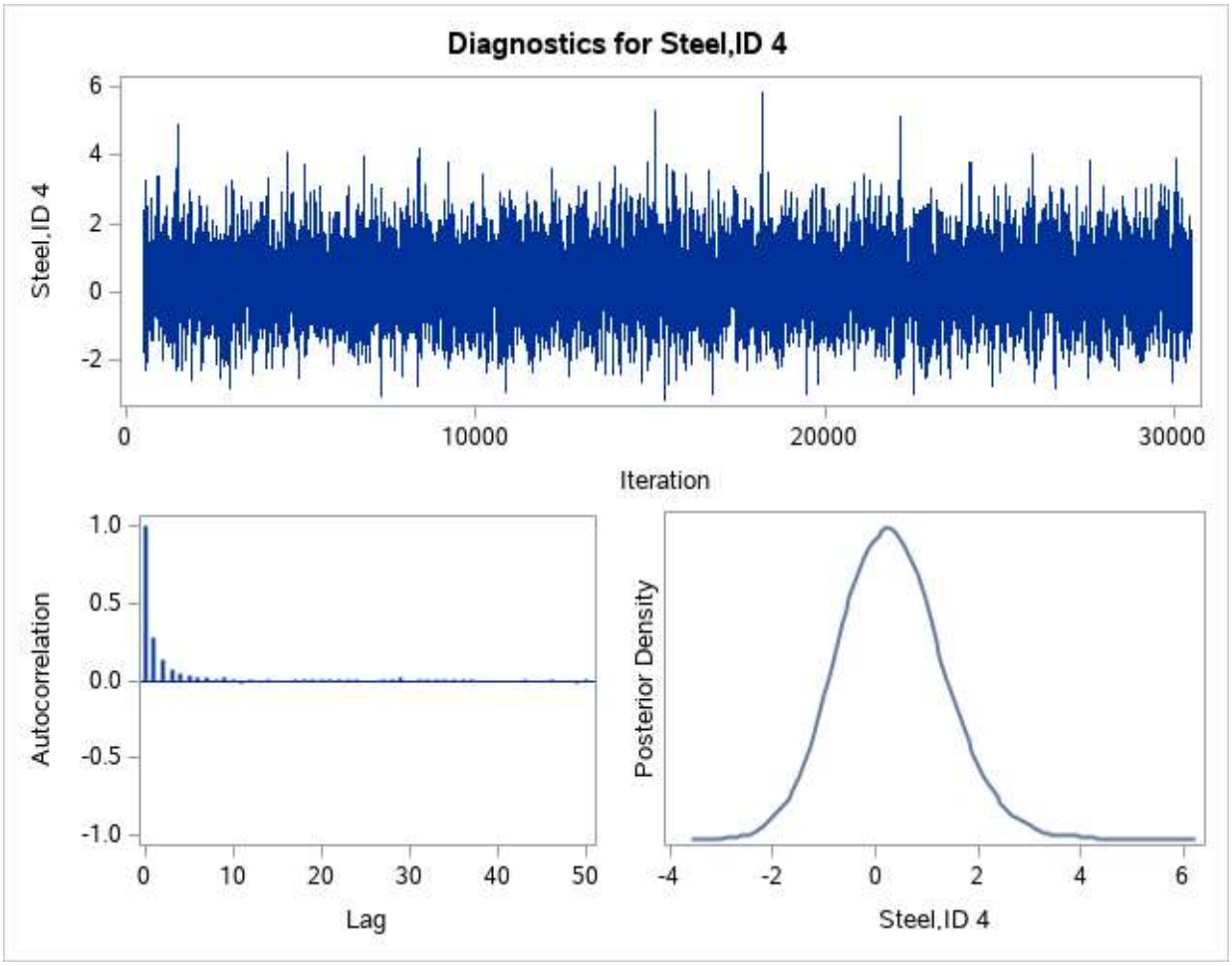

Diagnostics for Touchless,ID 3
Touchless,ID 3
2
0
-2
-4
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-4
-2
0
2
Touchless,ID 3

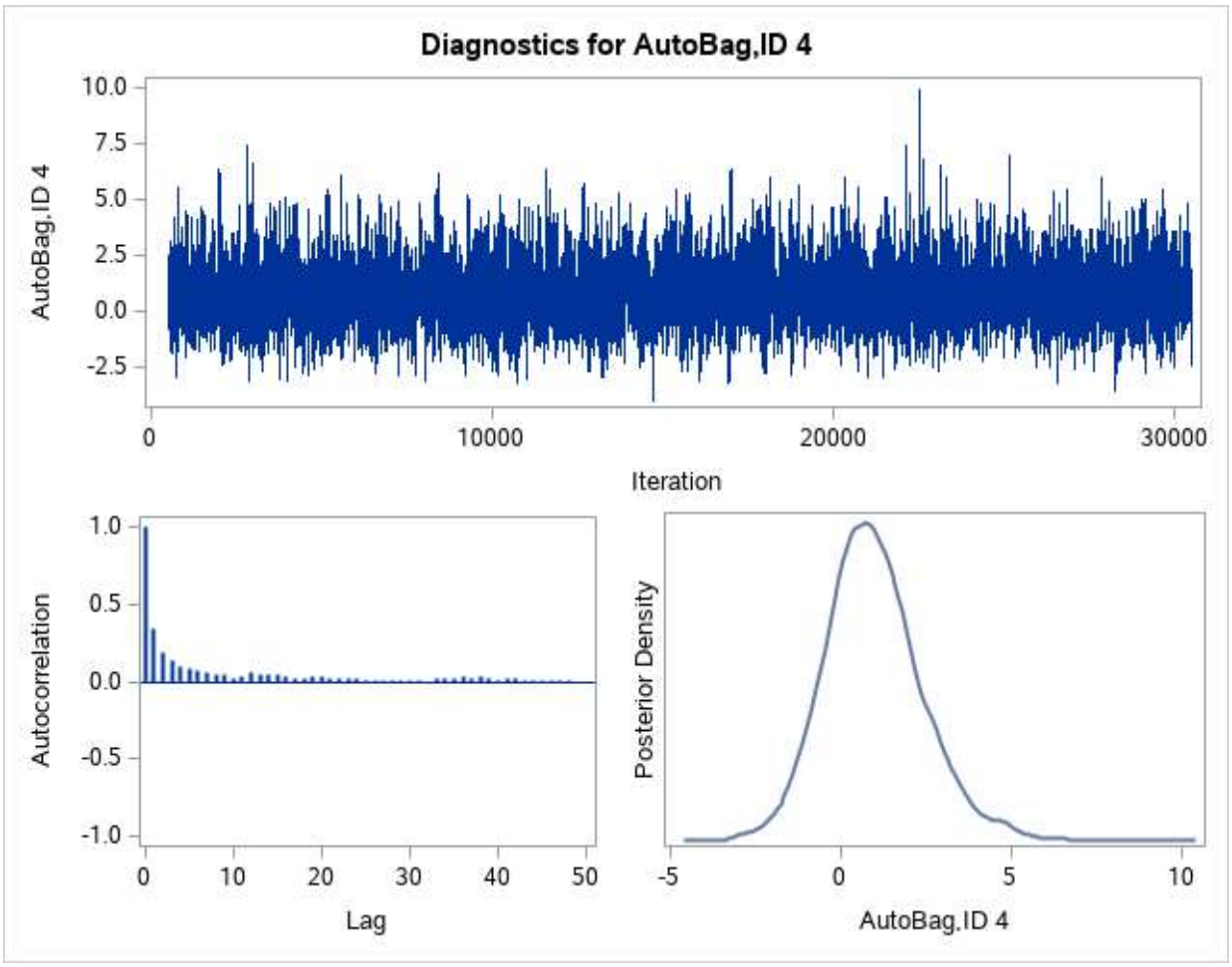

Diagnostics for Steel,ID 3
Steel,ID 3
4
2
0
-2
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-2
0
2
4
Steel,ID 3

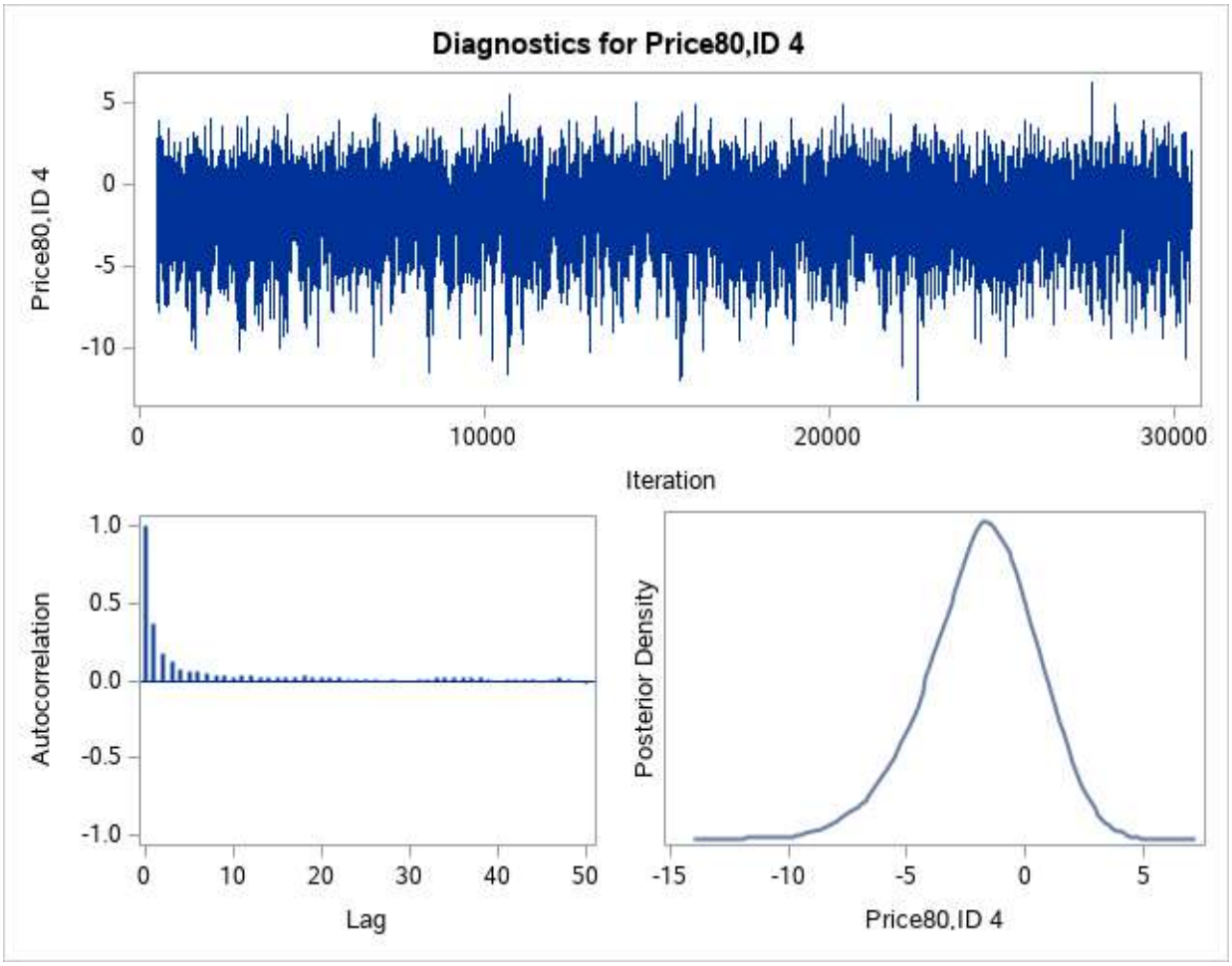

Diagnostics for AutoBag,ID 3
AutoBag,ID 3
2
0
-2
-4
-6
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-8
-6
-4
-2
0
2
4
AutoBag,ID 3

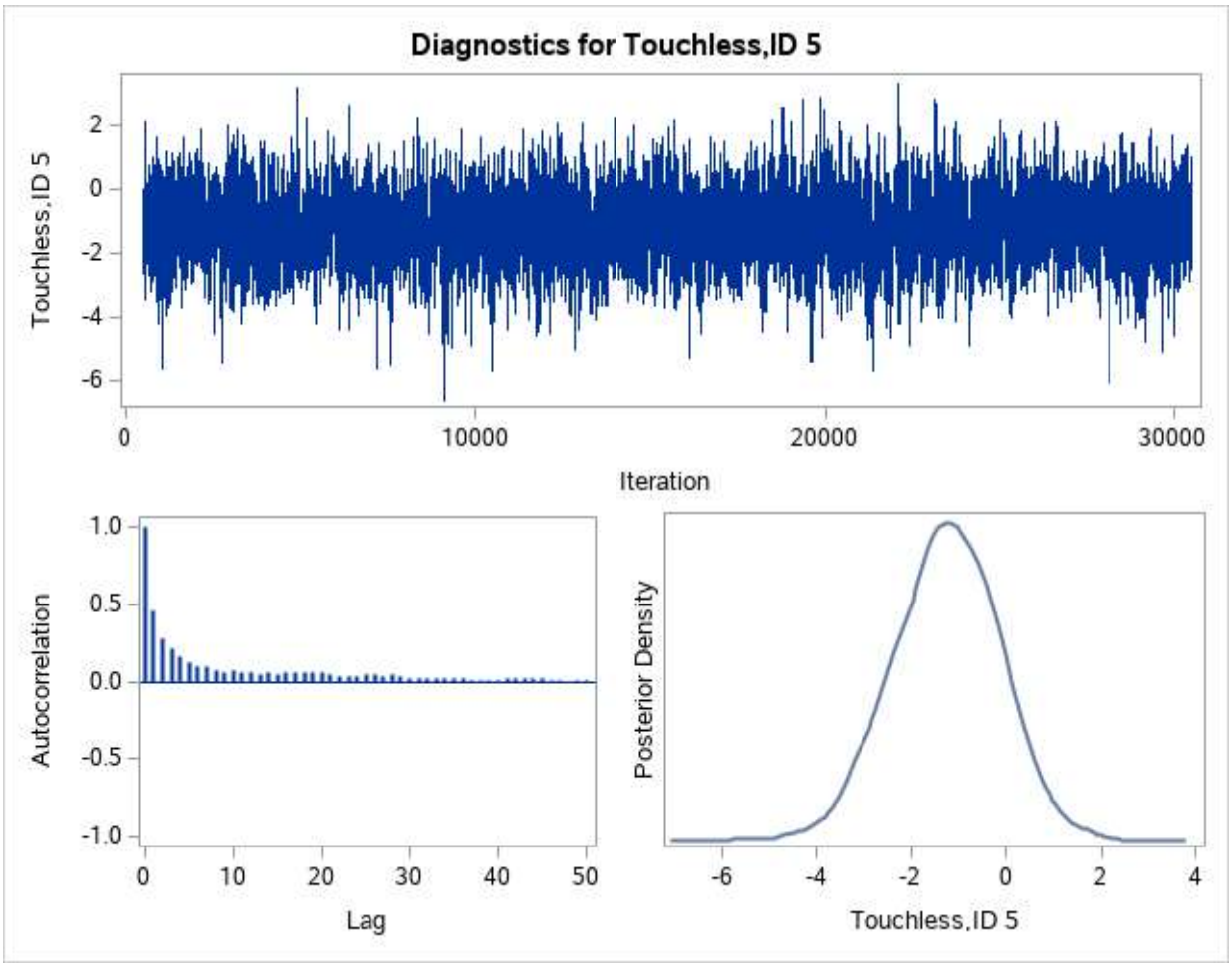

Diagnostics for Price80,ID 3
Price80,ID 3
10
5
0
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
0
5
10
Price80,ID 3

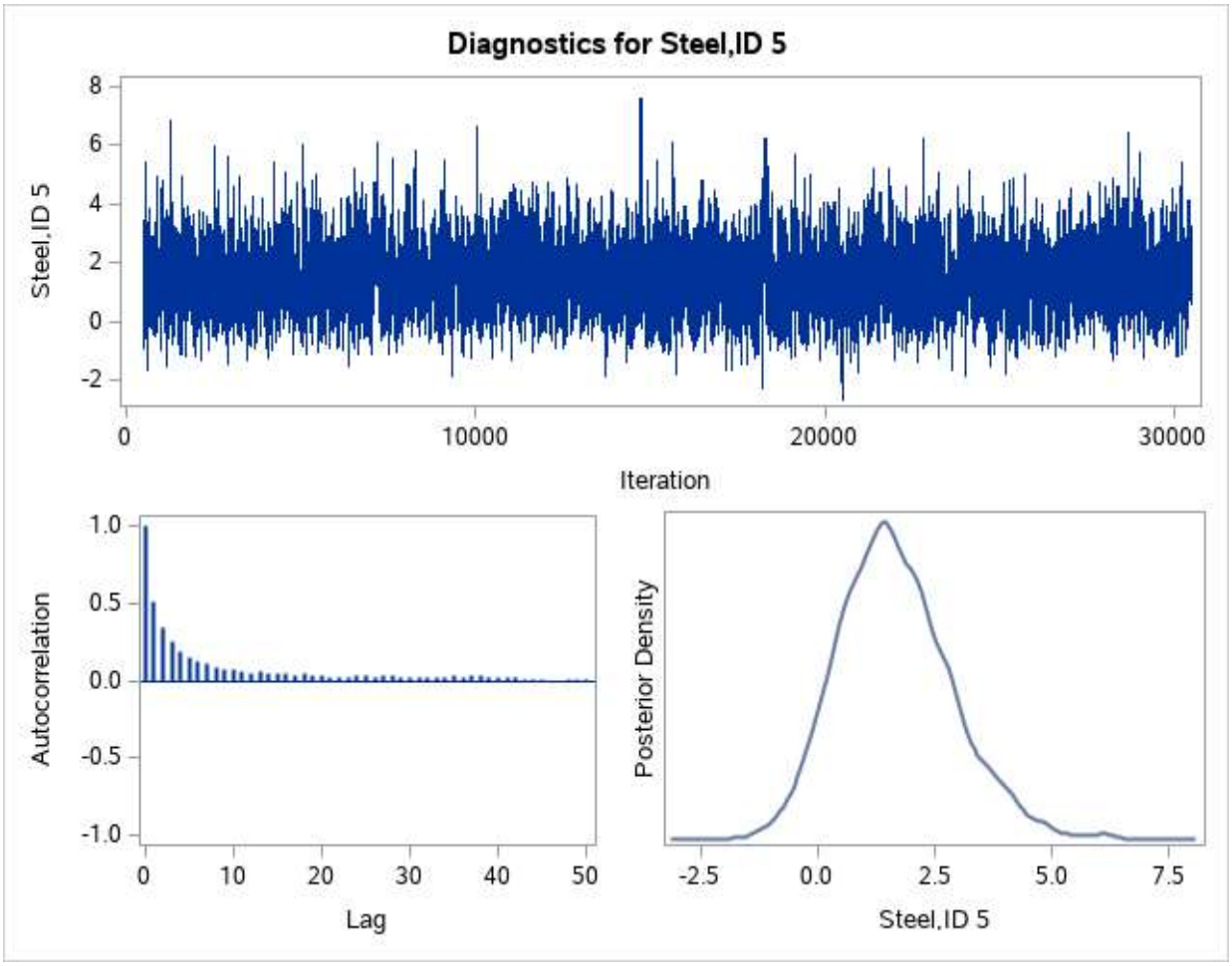

Diagnostics for Touchless,ID 4
Touchless,ID 4
2
0
-2
-4
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-4
-2
0
2
4
Touchless,ID 4

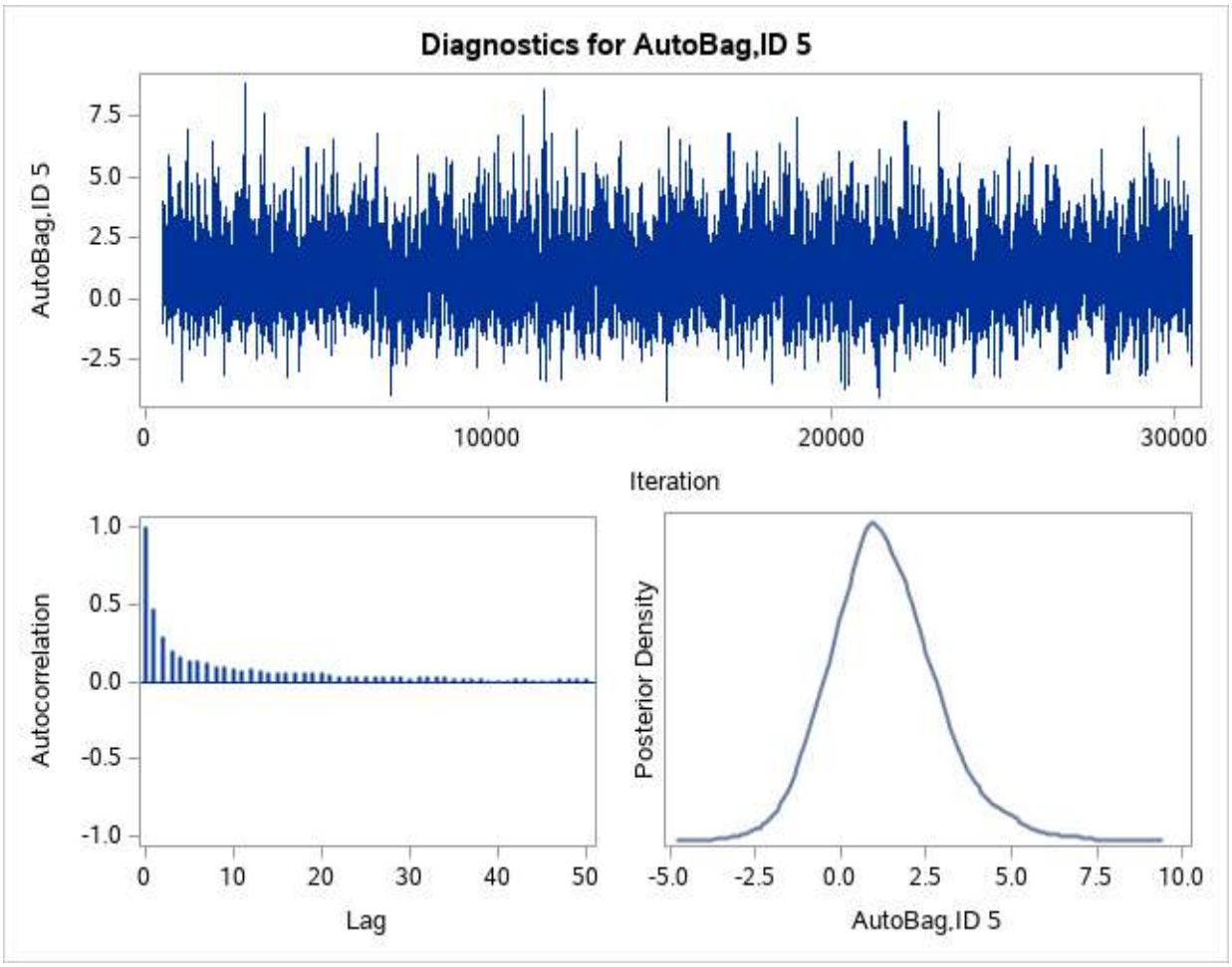

Diagnostics for Steel,ID 4
Steel,ID 4
Iteration
Autocorrelation
Lag
Posterior Density
Steel,ID 4

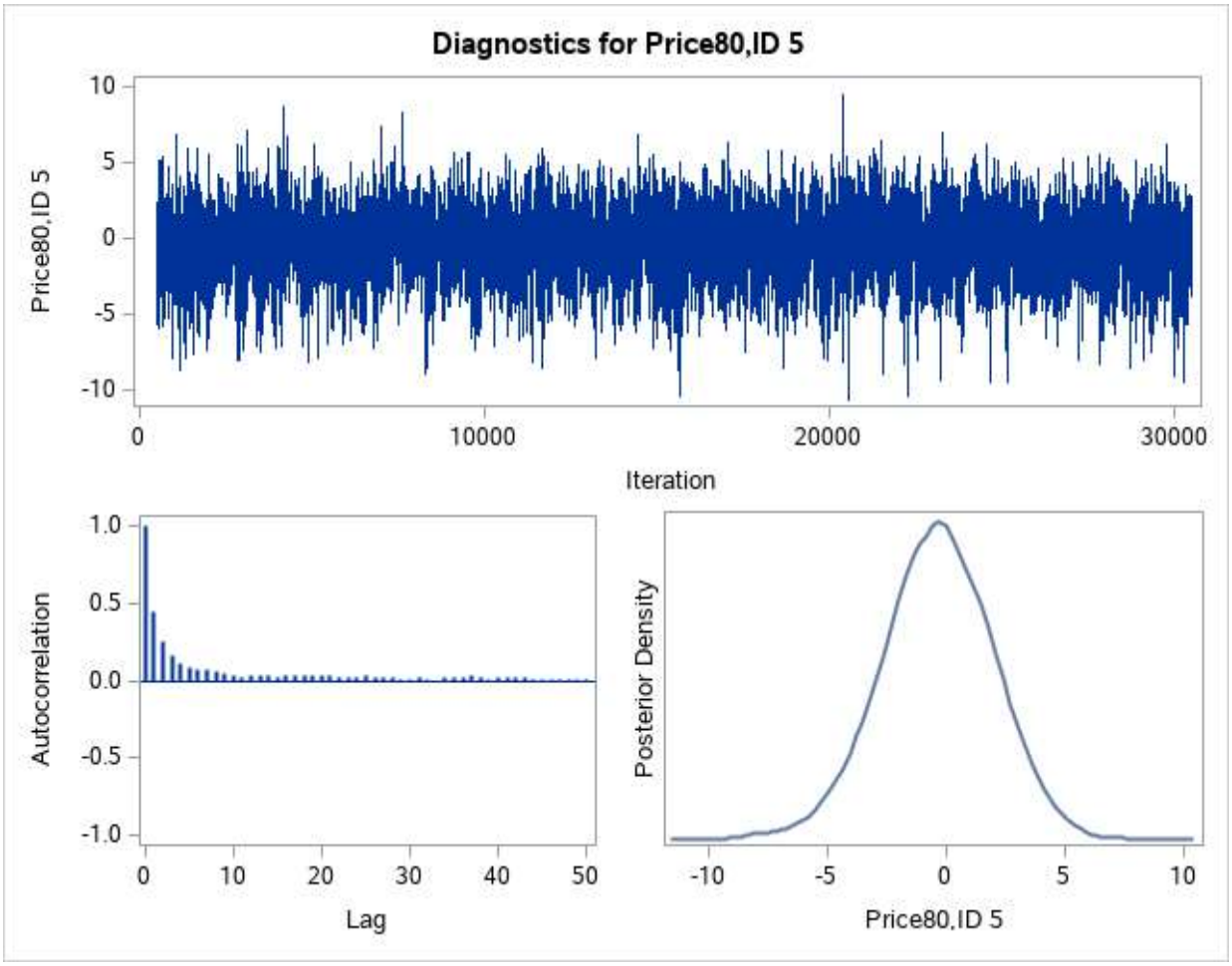

Diagnostics for AutoBag,ID 4
AutoBag,ID 4
Iteration
Autocorrelation
Lag
Posterior Density
AutoBag,ID 4


Diagnostics for Price80,ID 4
Price80,ID 4
5
0
-5
-10
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-15
-10
-5
0
5
Price80,ID 4


Diagnostics for Touchless,ID 5
Touchless,ID 5
2
0
-2
-4
-6
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-6
-4
-2
0
2
4
Touchless,ID 5


Diagnostics for Steel,ID 5
Steel,ID 5
8
6
4
2
0
-2
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-2.5
0.0
2.5
5.0
7.5
Steel,ID 5


Diagnostics for AutoBag,ID 5
AutoBag,ID 5
7.5
5.0
2.5
0.0
-2.5
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-5.0
-2.5
0.0
2.5
5.0
7.5
10.0
AutoBag,ID 5


Diagnostics for Price80,ID 5
Price80,ID 5
10
5
0
-5
-10
0
10000
20000
30000
Iteration
Autocorrelation
1.0
0.5
0.0
-0.5
-1.0
0
10
20
30
40
50
Lag
Posterior Density
-10
-5
0
5
10
Price80,ID 5

In [ ]: